



# Dynamic joint sensor selection and maintenance optimization in partially observable deteriorating systems

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## ABSTRACT

We consider a degrading system with costly but silent failures. The system can be partially observed using a set of heterogeneous noisy sensors at a given cost, where each sensor noise level is a function of the system state. The goal is to devise a dynamic joint sensor selection and maintenance optimization model to minimize the total expected discounted cost of the system. We develop an infinite-horizon discrete-time partially observable Markov decision process model to dynamically prescribe the sensor subsets for system monitoring and the timing of maintenance activities. We perform numerical experiments for a system with up to ten sensors and compare the proposed optimal policy with four heuristic policies. We observe that the proposed approach consistently outperforms all heuristic policies considered. Through a comprehensive analysis of numerical experiments, our study shows the efficacy of the proposed model's optimal policy in strategically selecting sensors to achieve precise system state estimation while managing costs. Through these experiments, we find that it is optimal to use lower-cost, higher-variance sensors and often fuse them for more accurate estimations. However, the choice depends on the system state and sensor quality distribution. Importantly, our model excels in determining the best action by considering these factors, underscoring its practical applicability.

## 1. Introduction

Condition-based maintenance (CBM) is a strategy that schedules maintenance activities based on the asset condition (Ahmad & Kamaruddin, 2012; Cai et al., 2022; Jardine, Lin, & Banjevic, 2006). Often, these activities are scheduled through periodic and/or continuous monitoring of specific measures of a system. Once these measures are considered to be unfavorable (e.g., one or more measurements exceeding a predefined threshold), a maintenance action is performed (de Jonge & Scarf, 2020; Peng, 2021; Tsang, 1995). System condition monitoring generally involves observing degradation-based measures, such as temperature, vibration, and pressure, to characterize the system health state. The monitoring may involve full inspections or sensor-based measurements, providing full or partial information about the system condition. The actions that are followed may include repair, further inspection, or replacement (Akçay, 2021; Qiu, Ming, Sallak, & Lu, 2021; Tsang, 1995). Overall, CBM is considered to be the most advanced state-of-the-art maintenance program applied in practice (Jardine et al., 2006; Kim & Makis, 2013). Extensive research findings

consistently support the notion that effective CBM can lead to notable enhancements in equipment reliability, all while ensuring significant cost reductions (Alaswad & Xiang, 2017).

With advances in sensing and computing, low-cost sensors are being widely developed and deployed to facilitate CBM (Bogue, 2013; Krishnamurthy et al., 2005; Larrañaga, Salokangas, Saarela, & Kaarmila, 2020). As such, a single asset may be monitored using an array of redundant or complementary sensors, where the data collected from the sensors are fused to provide a more accurate picture of the asset condition than that could be achieved by a single sensor (Klein, 1999). Data/sensor fusion is not, however, trivial as in practice, depending on the fusion method, the outputs can be misleading and even result in poorer outcomes than that achieved by considering only a subset of sensors (Hall & Llinas, 1997).

Sensor selection is the practice of selecting the most appropriate sensors for a particular application (Li, Wang, & Wang, 2017). In a broad sense, this may involve *a priori* selection of the most appropriate sensors based on availability, cost, accuracy, and basic operation conditioning requirements, among others. However, this task can be complicated if

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there is a dependency between system/operating condition and sensing requirements/accuracy, giving rise to the need for *dynamic* sensor selection. Dynamic sensor selection is the practice of selecting different sensors/sensing modalities in real time to achieve a specific goal under certain requirements and/or restrictions.

Dynamic sensor selection (and fusion) is not yet widely explored in the context of CBM. Dynamic sensor selection in the context of CBM may assist the decision maker to use the most informative set of sensors in order to better estimate the (partially observable) state of a deteriorating system. CBM has a wide variety of applications in manufacturing plants, nuclear power plants (NPPs), process industry, military (army, navy, air force) ground vehicles, IT infrastructure, commercial vehicles, and aviation/aircraft (Prajapati, Bechtel, & Ganesan, 2012). In CBM, the wired and/or wireless techniques can be used for data acquisition (Prajapati et al., 2012). Compared with wired monitoring systems, wireless monitoring systems have constrained resources including limited radio bandwidth, computational ability, and battery energy (Hou & Bergmann, 2012). The importance of considering power and bandwidth restrictions in real-time wireless condition monitoring has been widely discussed and established (Hou & Bergmann, 2012; Tang, Wang, Cattley, Gu, & Ball, 2018; Yilin, Ambrosino, & Sinopoli, 2011). Similarly, networks such as body sensor networks, or IoT sensors and devices are subject to bandwidth considerations (e.g., Guegan & Orgerie, 2019; Nabar, Walling, & Poovendran, 2010; Saghafian, Tomlin, & Biller, 2021). Indeed, the need for dynamic sensor selection often arises in CBM problems due to limited bandwidth or requisite processing power (e.g., Spaan & Lima, 2009), where only a limited number of sensors may be used for system monitoring.

In addition, although in general low-cost sensors are widely available in certain use cases, that is not necessarily the case in all contexts, e.g., in military applications or in NPPs where sensors must meet certain safety requirements, among others (Rajinder, Sankaranarayanan, & Harvel, 2021). Furthermore, while sensors themselves may not incur direct costs for data collection, there is a significant consideration for data processing costs associated with different sensor types. For example, processing video data may be more expensive for engineers or operators compared to processing vibration sensor data, despite the former being more accurate. In addition, in certain systems, not all sensors present the same processing/observation costs and/or accuracy under different operating conditions of a system. For example, in an aircraft, the inertial sensor measurements become less accurate (due to sensor drift) as the aircraft descends, while the instrument landing system and radio altimeter sensor measurements become more accurate (Wiegman, 2019). As another example, vibration in turbines, pumps, or compressors can impact the accuracy of thermocouple temperature readings. This is due to vibrations-induced material fatigue in the thermocouples and the consequent insulation failure (Jethra, 2018). Thus, under such conditions, dynamically adjusting which sensors to use may provide a more accurate estimate of the system condition.

Practical examples where dynamic sensor selection is applicable include wind turbine maintenance and pipeline integrity management. In the case of wind turbines, various sensors can be deployed to monitor parameters such as wind speed, blade vibration, gearbox temperature, and generator performance. By selecting and fusing data from appropriate sensors, maintenance personnel can obtain a comprehensive understanding of the turbine's health condition. For instance, if the wind speed is low and the turbine is operating under normal conditions, some sensors may be put into sleep (no observing) mode to conserve power. However, if a critical fault is detected or the wind speed increases, the sensors can be activated to provide real-time monitoring and facilitate prompt maintenance actions (Wang & Ma, 2012; Zhang & Ma, 2016). Similarly, in pipeline integrity management, sensors are deployed along the pipeline network to monitor its condition. Through dynamic sensor selection, the most appropriate subset of sensors can be chosen for each pipeline segment, considering their degradation characteristics, to improve monitoring accuracy and enable informed

maintenance decisions (Cardell-Oliver, Scott, Chapman, Morgan, & Simpson, 2015; Kim, Sharma, Boudriga, & Iyengar, 2010; Saeed, Ali, Rashid, Qaisar, & Felemban, 2014).

Other domains where dynamic sensor selection is relevant include wireless structural health monitoring. Consider a wireless sensor network (WSN) deployed for structural health monitoring of a large infrastructure, such as a bridge or a building. The WSN consists of a network of wireless sensors distributed across the structure to measure parameters like vibrations, strain, temperature, and humidity. The objective is to detect and predict structural abnormalities or potential failures. In this scenario, sensor selection and control sleep play crucial roles in optimizing the network's performance and energy efficiency. The WSN consists of a diverse set of sensors, each with different capabilities, costs, and power requirements. Depending on the specific monitoring requirements and the criticality of different structural components, sensor selection can be dynamically adjusted to optimize resource allocation and improve monitoring accuracy. For example, certain sensors may be more suitable for detecting vibrations, while others may be more accurate in measuring strain. By selecting and activating the most relevant sensors based on the structural behavior and monitoring objectives, the network can achieve a higher level of accuracy and reduce unnecessary data collection (Fuemmeler, Atia, & Veeravalli, 2011; Weimer, Sinopoli, & Krogh, 2008).

In this study, our goal is to address a specific problem in CBM by leveraging dynamic sensor selection and fusion techniques within the context of sequential decision making under uncertainty and partial information. We focus on, a stochastically degrading system, whose true status is unknown and can only be partially observed using heterogeneous noisy sensor(s). The system is subject to failure. Once failed, a failure cost is incurred in every period. However, system failure is silent and may only be detected through (noisy) sensor observations. At each decision epoch, a subset of sensors may be used to observe the system condition. We assume that for each sensor, the measurement accuracy is a function of the system state. The collected data are then fused, when more than one sensor is used for monitoring, to estimate the system state. We also allow the number of sensors used for system monitoring to be restricted due to system bandwidth limits. Additionally, we assume that sensor data observation is costly, and the cost associated with each sensor is a function of its overall accuracy. Therefore, there exists a trade-off between monitoring the system to improve the accuracy of estimating its state and replace it when it is considered to be in the failed state, and foregoing either of these actions at the risk of incurring a silent failure cost. Our research aims to develop a comprehensive approach that combines dynamic sensor selection and fusion as well as system maintenance to estimate the system state and minimize the total expected discounted cost over an infinite horizon. By considering the dynamic selection of sensors, the reliability of sensor measurements in relation to the system condition, and the trade-off between system monitoring and maintenance, we seek to provide insights into effective CBM strategies in real-world scenarios.

Our study touches on and builds upon works in the literature on CBM, dynamic sensor selection and fusion, and maintenance optimization. Due to the challenges of the problem of interest, it provides unique perspectives and contributions. Our major contributions are as follows. This study integrates sensor scheduling/selection into CBM in order to minimize the total expected costs of the system. In addition, it accounts for sensor noise and the dynamically changing sensor accuracy as a function of system state. Lastly, this study investigates the impact of using only a restricted subset of sensors for monitoring, and draws insights as to when sensor fusion may be beneficial. This potentially enables handling more complex and practical CBM problems in the future.

The remainder of the paper is organized as follows. In Section 2, we provide a summary of the existing related works. In Section 3, we present the mathematical formulation. Next, in Section 4, we first present the results of numerical experiments for a system with two

candidate sensors to draw insights and then conduct expanded numerical experiments for a large-scale ten-sensor system to capture a more practical setting and provide comparisons with heuristic policies. Finally, we conclude in Section 5.

## 2. Literature review

### 2.1. Maintenance in partially observable deteriorating systems

The joint decision on optimal timing of (costly) observations and scheduling maintenance activities has been studied in the literature. For instance, [Yeh \(2003\)](#) considered a partially observable deteriorating system whose true state could only be revealed by imperfect inspections. Under the long-run average cost criterion, the author analyzed the inspection-repair-replacement (IRR) policy. [Maillart \(2006\)](#) examined the problem of adaptively scheduling observations (both perfect and imperfect) and preventive maintenance actions for a deterioration system with obvious failures such that the long-run average-cost per unit time is minimized. [Elwany, Gebraeel, and Maillart \(2011\)](#) developed a replacement decision model that used in degradation signals from a partially degrading component to determine the optimal replacement time. [Kim and Makis \(2013\)](#) addressed the problem of joint optimization of imperfect system condition sampling and preventive maintenance in a system with obvious failures to minimize the long-run expected average cost per unit time. [Chen, Ye, Xiang, and Zhang \(2015\)](#) addressed the problem of optimal inspection and replacement policy of a deteriorating system with silent failures whose degradation is modeled as an inverse Gamma process. They assumed system failures could only be detected through perfect and costly observations, with the goal of minimizing the total operational costs. [Karabağ, Eruguz, and Basten \(2020\)](#) addressed the challenges of monitoring multiple critical components in advanced technical systems where installing dedicated sensors for each component is not practical. It introduced a CBM policy that works with partial information about the system's condition and employed an infinite horizon POMDP model to make informed maintenance decisions. Similarly, [Kıvanç, Özgür Ünlüakın, and Bilgiç \(2022\)](#) focused on maintenance optimization for multi-component dependent systems, which presents a challenging problem. This work employed factored POMDPs as a powerful tool to address the uncertainty inherent in stochastic environments. [Zhang and Zhang \(2022\)](#) focused on the problem of optimal inspection and replacement policy for a deteriorating system in which the machine deteriorates according to a Markov process until it fails. This study presented an exact, analytical solution using a dual framework for POMDP model. [Zhao and Smidts \(2022\)](#) addressed the problem of optimizing maintenance policies under imperfect knowledge of system degradation models and partial observability of degradation states. The proposed approach utilized a reinforcement learning method specifically designed for maintenance problems with Markov degradation processes. [Roux, Fang, and Barros \(2022\)](#) presented a novel approach to address a maintenance planning problem using a POMDP model. The focus of this study was on a single-unit system that undergoes random degradation and failures, with its current degradation state being partially observed through an imperfect monitoring system. [Morato, Papakonstantinou, Andriotis, Nielsen, and Rigo \(2022\)](#) introduced a comprehensive approach that integrates dynamic Bayesian networks and POMDPs to address the optimization of inspection and maintenance planning. This study focused on the context of structural reliability and provided the necessary formulations for developing both infinite and finite horizon POMDPs. [Wari, Zhu, and Lim \(2023\)](#) addressed the problem of corrosion-induced failure in pipelines used for transporting oil and gas products. To optimize maintenance operations and mitigate the impact of corrosion, the paper proposed a POMDP model. The model utilized inline inspection to monitor the corrosion progress and assess the extent of pipeline deterioration. While there may be certain superficial resemblances between our study and previous research on maintenance in deteriorating systems using POMDP models, it is important to note that none of these studies specifically address the critical aspect of sensor selection.

### 2.2. Sensor selection in CBM

In the CBM literature, there are several studies that use a set of sensors/observations (as opposed to a single sensor) for the system condition monitoring. For instance, [Lin \(2014\)](#) derived optimal Bayesian maintenance policy for a system in which multiple vibration sensors were used to monitor the system at predefined equidistant sampling epochs. They assumed that vibration data were collected from different sensors and contained different partial information about the system and derived multivariate Bayesian control charts to minimize the long-run expected average cost per unit time. [Wang \(2012\)](#) developed a simulation-based multivariate Bayesian control chart for CBM of a partially observable multi-state system with obvious failures. In this study, samples from all sensors were collected at regular time epochs at a cost and a full inspection and possible repair occurred if the probability of the system being in a defective state is above a pre-defined threshold. [Liansheng, Wang, Liu, Zhang, and Peng \(2015\)](#) proposed an entropy-based approach to solve the sensor selection problem for condition monitoring and prognostics of aircraft engines. [Changyue and Liu \(2018\)](#) designed a statistical degradation model by combining extracted features and sensor signals where sensors were selected in a preprocessing step. Similarly, [Hiruta, Uchida, Yuda, and Umeda \(2019\)](#) proposed a design method of data analytics processes and an engineering tool for a CBM problem. As part of the analysis, this study selected sensors in a preprocessing step. Consequently, [Changyue, Liu, and Zhang \(2019\)](#) proposed a generic framework to solve the sensor selection problem for multi-sensor degradation modeling in CBM. Our study shares similarities with these highlighted studies in the use of multiple sensors in CBM. However, different from these studies, we consider the optimal timing of observations through dynamic sensor selection, i.e., not all sensors are used at predefined sampling epochs.

### 2.3. Dynamic sensor selection in CBM

With the ever increasing use of sensors in asset monitoring, there is a timely need to investigate the joint dynamic sensor selection and maintenance planning decisions in CBM; however, the current literature on CBM mostly involves the use of a *single* sensor or *all* sensors ([Ahmad & Kamaruddin, 2012](#); [Jardine et al., 2006](#); [van Oosterom, Maillart, & Kharoufeh, 2017](#)). To the best of our knowledge, there is only one study that considers dynamic sensor selection in the context of CBM. [van Staden and Boute \(2021\)](#) developed a CBM policy for a deteriorating system whose status can be monitored using an internal sensor or a more accurate, but costly external sensor. The decision maker then decides whether to continue system operation with the internal sensor, collect data from the external sensor, or perform maintenance. The authors also extended their model for a case where more than one external sensor was available for system monitoring. In contrast to our study, in this study, sensor accuracy is assumed to be fixed (i.e., it is not a function of system state) and system failure is outwardly observable. These sizable differences in problem settings result in the need of new methods to address our problem of interest.

Among the studies that use a single sensor to monitor a system, [van Oosterom et al. \(2017\)](#) most closely relates to our work. [van Oosterom et al. \(2017\)](#) investigated the problem of optimally maintaining a safety-critical system and its imperfect deteriorating sensor. The policy adaptively schedules full inspections and replacement of the sensor. They developed a POMDP model with the objective of minimizing the total expected discounted costs associated with system operation, full inspection, system replacement, and sensor replacement. While our study does not directly consider deteriorating sensors, the change in sensor accuracy as a function of true system state produces comparable dynamics and challenges. This work is, however, ultimately distinct, as in this study, we also account for multiple sensors and dynamic sensor selection.

### 3. Model formulation

In this section, our objective is to formulate a POMDP model that integrates dynamic sensor selection and maintenance optimization for a deteriorating system. The POMDP framework enables us to address the challenges posed by the uncertainty and partial observability of the system's true core state, as well as the need for sequential decision making under these conditions. By incorporating dynamic sensor selection and maintenance decision into the model, we aim to develop a comprehensive approach that minimizes the total expected discounted cost over an infinite horizon. In the subsequent sections, we proceed to establish the foundation of our study by defining the underlying dynamics of the system. This entails a comprehensive description of the core states, observations, and transition probabilities within the system. Furthermore, we present a comprehensive exposition on the details pertaining to actions, the associated costs, and the formulation of the POMDP model. Finally, we provide a concise overview of the heuristic policies considered for the purpose of comparative analysis followed by a description of the model extension to a system with multiple machine.

#### 3.1. Core state

Consider a system that undergoes multistate stochastic deterioration according to a time homogeneous, discrete-time Markov chain. Let  $s \in S = \{0, 1, \dots, \gamma + 1\}$  denote the (hidden) core state of the system, where states  $s = 0$  and  $s = \gamma + 1$  denote the brand new and failed states, respectively. In addition, states  $1, \dots, \gamma$  represent the intermittent system states in an increasing deterioration order. Note that the system core state is not outwardly observable and can only be partially observed using noisy observations. Therefore, the system belief state is used to present the probability that the system occupies state  $s$  at time  $t$ . Let  $\pi_t = [\pi_t(0), \pi_t(1), \dots, \pi_t(\gamma + 1)]$  denote the system belief state at time  $t$ , where  $\pi_t(s)$  is the probability that the system is in core state  $s$  at the beginning of period  $t$ . Note that  $\sum_{s \in S} \pi_t(s) = 1$ .

#### 3.2. Action

Assume that there are  $D$  heterogeneous candidate sensors for system monitoring. Let  $a_i$  indicate whether sensor  $i$ ,  $i = 1, \dots, D$ , is used for system monitoring ( $a_i = 1$ ) or not ( $a_i = 0$ ) at a given time epoch. Therefore, there are  $2^D - 1$  different sensor combinations in which at least one sensor is used for system monitoring. Let  $a$  denote the action taken at a given epoch. In cases where at least one sensor is used for monitoring, we define a one-to-one mapping that presents action  $a$  as a function of the subset of candidate sensors used. Specifically, we define  $a = g(a_1, a_2, \dots, a_D) = \sum_{i: a_i=1} 2^{i-1}$ , when  $1 \leq \sum_i a_i \leq D$ . Additionally, we may choose to 'do nothing' (forgo system monitoring) or replace the system under which cases the corresponding actions are denoted by  $a = 0$  and  $a = 2^D$ , respectively. Thus, we denote the set of actions by  $A = \{0, 1, 2, \dots, 2^D\}$ . For example, in a system with  $D = 2$  sensors, if we only use sensor 1 (i.e.,  $a_1 = 1$  and  $a_2 = 0$ ), the corresponding action is  $a = 1$ , whereas, if we only use sensor 2 or both sensors, the corresponding actions are  $a = 2$ , and  $a = 3$ , respectively. Finally, actions  $a = 0$  and  $a = 4$  correspond to 'do nothing' and replacement, respectively. That is,  $A = \{0, 1, 2, 3, 4\}$ . Additionally, we assume that the number of sensors to be used at each time epoch might be restricted by the system bandwidth. We denote the restricted action space by  $A_K^{\mathcal{R}}$  when the number of sensors to be used is limited to  $K$  sensors at a time. In the example with  $D = 2$  candidate sensors, if the number of sensors to be used is limited to one sensor at a time, we have  $A_1^{\mathcal{R}} = \{0, 1, 2, 4\}$ .

Although the current formulation focuses on dynamic sensor selection, it is important to note that maintenance optimization is an integral part of the proposed model. In this model, maintenance refers to the action of system replacement. We explicitly include maintenance decisions by incorporating system replacement as one of the available actions for the decision-maker. The objective of maintenance

optimization in our model is to determine the optimal timing for system replacements based on observed sensor data and estimated system state. By jointly optimizing sensor selection and maintenance actions, our model captures the trade-off between monitoring the system, improving state estimation accuracy, and making informed decisions on system replacements. It is important to note that maintenance (represented by system replacement in our model) plays a crucial role in mitigating the risk of silent failures and minimizing the associated costs that is a one of the main CBM promises.

#### 3.3. Sensor observations

The  $D$  candidate sensors used for system monitoring are assumed to be heterogeneous, where each sensors' accuracy is a function of the system state. That is, due to sensor design and/or the physical property they measure, any given sensor can present various degrees of accuracies in different system states. For instance, a sensor that measures vibrations may be more accurate in a more deteriorated system state. Additionally, sensor data are assumed to be conditionally independent. That is, given the system core state, observations made using one sensor are not affected by those obtained using another sensor. This assumption simplifies the modeling process and allows us to focus on individual sensor observations without considering complex inter-sensor dependencies.

We assume sensor observation is a normal random variable whose variance in different system core states are previously established through testing. In addition, we assume that in any system core state, the corresponding system features (e.g., temperature, pressure, etc.) have some inherent variability. That is, feature  $f$  is a normal random variable with known mean and variance that are functions of system core state  $s$ . However, for notation simplicity, we assume all feature measurements are adjusted (e.g., a shifted distribution) such that the system means in any given state simply collapse to single numbers. In addition, we assume the sensor noise and inherent system variance in any state are independent. Let vector  $\mu = [\mu_0, \dots, \mu_{\gamma+1}]$  denote the system mean in different system core states, i.e.,  $\mu_s$  specifically denotes the overall mean of the system when the system is in core state  $s \in S$ . Moreover, let  $\sigma_i^2 = [\sigma_i^2(0), \sigma_i^2(1), \dots, \sigma_i^2(\gamma + 1)]$  denote the variance when using sensor  $i$ , where  $\sigma_i^2(s)$  is the overall corresponding variance in system state  $s \in S$ . We assume that in any data collection instance, a sample of size  $n$  is drawn from each selected sensor. The sample size  $n$  is assumed to be fixed and derived based on sensor specifications (sampling frequency). This sample size is considered as the sampling frequency in one epoch and is treated as an input to the model. While we assume a fixed sample size across sensors, we acknowledge that it could be variable without impacting the model. The impact of  $n$  is primarily on the observation probabilities (information matrix) and does not significantly affect the model itself. Regarding the calculation of information matrices, we assume them as given parameters. If raw data is available, existing algorithms can be employed to infer these parameters. Let  $\bar{X}_i^s$  denote the sample mean of the data collected from sensor  $i$  when the system is in state  $s$ . Therefore,  $\bar{X}_i^s \sim \mathcal{N}\left(\mu_s, \frac{\sigma_i^2(s)}{n}\right)$ ,  $s \in S$ .

If more than one sensor is used to collect data, the observations will be fused. Let  $\tilde{X}_a^s$  denote the sample mean of the data collected from sensors under action  $a \in \{1, 2, \dots, 2^D - 1\}$  when the system is in state  $s \in S$ . Per Fraser-Potter fixed-interval smoother (Maybeck, 1982), the random variable  $\tilde{X}_a^s$  is derived as the weighted average of the selected sensors' sample means, where the weights are the inverses of the corresponding variances of sample means, i.e.,

$$\tilde{X}_a^s = \frac{\sum_{i: a_i=1} \frac{n}{\sigma_i^2(s)} \cdot \bar{X}_i^s}{\sum_{i: a_i=1} \frac{n}{\sigma_i^2(s)}}, \quad \forall s \in S. \quad (1)$$



Clearly,  $\tilde{X}_a^s$  is normally distributed with

$$E(\tilde{X}_a^s) = \mu_s, \quad \text{Var}(\tilde{X}_a^s) = \frac{1}{\sum_{i:a_i=1} \frac{n}{\sigma_i^2(s)}} = (\sigma_s^a)^2. \quad (2)$$

Note that  $\tilde{X}_a^s$  is the best linear unbiased estimator (black) (Xiao & Luo, 2005), and because of weighting the sensor sample means by the inverses of their corresponding variances, Eq. (1) never underperforms the system state estimation achieved by a single sensor. In addition, note that in a case where only a single sensor, sensor  $i$ , is used for data collection, we simply have  $\tilde{X}_a^s = \tilde{X}_i^s$ . By weighting the sensor sample means based on their variances, we effectively reduce the influence of sensors with higher variability in the fusion process. Sensors with higher variability will be assigned lower weights, which means their measurements contribute less to the overall estimation. This weighting scheme acts as a form of compensation, ensuring that the impact of sensor heterogeneity is mitigated during system state estimation.

The choice of assuming a normal distribution for sensor observations in our model serves as a practical approximation that balances simplicity and tractability. By assuming normality, we can leverage well-established statistical techniques and inference methods that are widely applicable and readily available. This allows for efficient estimation and fusion of sensor data. Importantly, the assumption of a normal distribution does not limit the model's applicability to only normally distributed data. Specifically, this assumption only impacts the estimation of observation probabilities (information matrix) in our model. In deed, the information matrix can also be estimated using existing algorithms such as Baum-Welch when raw data is available (Bilmes et al., 1998). The POMDP framework is flexible and can accommodate different types of observation distributions. If real data or prior knowledge suggest a different distribution, such as a skewed or heavy-tailed distribution, the model can be adapted accordingly without requiring an explicit reformulation of the entire framework.

Let  $o \in \mathcal{O} = S \cup \{-1\}$  denote the observed system state obtained from sensors data. Note that “observed” system state  $o$  might be different from system core state  $s$  due to sensors' inherent noise. If data are collected using sensors ( $a = \{1, \dots, 2^D - 1\}$ ), the observed system state can be obtained using sensors data. Otherwise, if no data are collected ( $a = \{0, 2^D\}$ ), for completeness, we let  $o = -1$ . Let  $Q_a$  denote the information matrix associated with action  $a$ . Specifically,  $Q_a(o|s)$  denotes the probability that the observed system state is  $o$ , given that the system is in the core state  $s$  and action  $a \in \{1, \dots, 2^D - 1\}$  is taken. Details on the estimation of information matrices are provided in Appendix A.

### 3.4. Transition probabilities

Let  $P$  denote the one-step transition probability matrix (TPM) of the system, where  $P(s'|s)$  denotes the probability that the system transitions to state  $s'$  from state  $s$  in one time epoch. As discussed, the system state is not fully observable, but it can be partially observed using sensor observations. As such, at each decision epoch, depending on the action taken, the system belief state can be updated. In case of no data collection ( $a = 0$ ), system's TPM is used to update the system belief. When the action is to collect data from one sensor or a combination of sensors, the system belief is updated using the observed system state  $o \in \mathcal{O}$  and by applying the Bayes' theorem. Let  $B_{t+1} = \beta[\pi_t, a, o]$  denote the updated belief at time  $t + 1$  starting from the system belief  $\pi_t$  at time  $t$ , under action  $a$ , and the observed system state  $o$ . Therefore, the updated belief that the system is in state  $s$  is given by

$$\beta[\pi_t, a, o](s) = \begin{cases} \sum_{j=0}^{Y+1} \pi_t(j)P(s|j), & \text{if } a = 0, \\ \frac{\sum_{j=0}^{Y+1} \pi_t(j)P(s|j)Q_a(o|s)}{\sum_{j'=0}^{Y+1} \sum_{j=0}^{Y+1} \pi_t(j)P(s'|j)Q_a(o|s')}, & \text{if } a \in \{1, \dots, 2^D - 1\}, \\ \mathbb{1}_{\{s=0\}}, & \text{if } a = 2^D, \end{cases} \quad (3)$$

where  $\mathbb{1}_{\{\cdot\}}$  is an indicator function. The first statement (when  $a = 0$ ) simply updates the belief by relying on the TPM. The second statement

(when  $a \in \{1, \dots, 2^D - 1\}$ ) updates the belief by relying on the TPM and further adjusting it per the likelihood of observing  $o$ . The denominator is simply a normalizing factor. Lastly, in the case of system replacement (when  $a = 2^D$ ), the system transitions to brand new state ( $s = 0$ ) with certainty. In this case, the updated belief is  $e_0$ , where  $e_0$  is the vector of all zeros with a one in the first position.

### 3.5. Costs

In our model, we consider a cost associated with processing the collected data whenever a fusion action is triggered, such as when the system dynamics indicate a potential deterioration state. Let  $c_i$ ,  $i = 1, \dots, D$ , denote the per period processing/observation cost of sensor  $i$  when sensor  $i$  is used for data collection. By incorporating the assumption of data observation costs, our model reflects the practical realities faced by organizations implementing CBM strategies. However, we acknowledge that this assumption introduces certain limitations to the applicability of the model, where data observation cost is not a significant constraint.

In addition, we assume that system replacement is costly. Let  $c_R$  denote the cost of system replacement. Although system monitoring and replacement are costly, foregoing these actions can have negative consequences. We assume system failure is silent (Ivy & Pollock, 2005; Maillart & Zheltova, 2007) and the cost of  $c_F$  is incurred for each epoch that the system remains in the failed state. Therefore, there exists a trade-off between performing monitoring to infer the system state and replacing the system when it is considered to be in the failed state, and foregoing these actions and incurring the cost of silent failure. Note that we assume data observation and replacement costs incur in the beginning of the period, and silent failure costs incur at the end of the period.

### 3.6. Optimality equations

The objective of the POMDP model is to minimize the total expected discounted cost over an infinite horizon. Let  $\lambda$  denote the discount factor and  $Y(\pi_t)$ ,  $W^a(\pi_t)$ , and  $Z(\pi_t)$  denote the total expected discounted cost-to-go starting from the system belief state  $\pi_t$  under actions  $a = 0$  (do nothing),  $a \in \{1, \dots, 2^D - 1\}$  (monitor the system), and  $a = 2^D$  (system replacement), respectively. Also, let  $V_t^*(\pi_t)$  denote the minimum total expected discounted cost-to-go starting from the system belief state  $\pi_t$ . Hence, per (Krishnamurthy, 2016), for  $a \in A$  we have

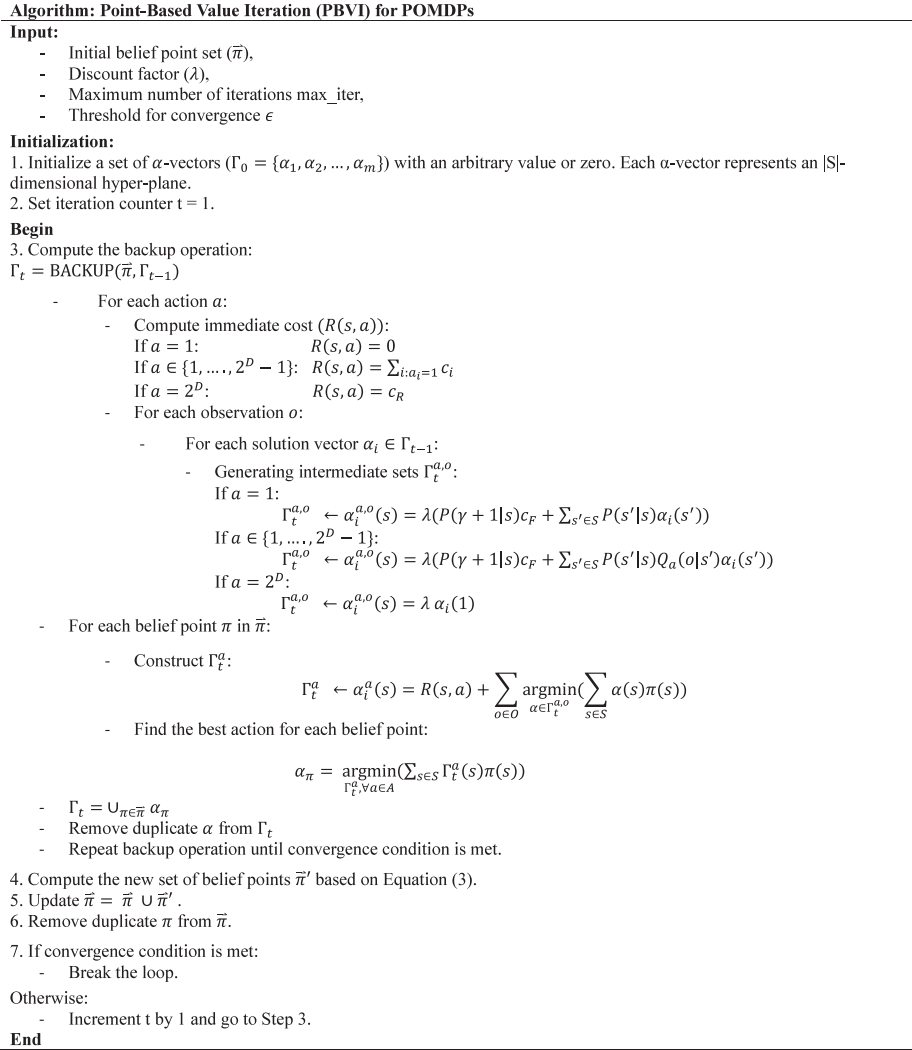
$$Y(\pi_t) = \sum_{s \in S} \pi_t(s) \left( 0 + \lambda \cdot \left[ P(\gamma + 1|s) \cdot c_F + \sum_{s' \in S} P(s'|s) V_{t+1}^*(\beta[\pi_t, 0, -1]) \right] \right), \quad (4)$$

$$W^a(\pi_t) = \sum_{s \in S} \pi_t(s) \left[ \sum_{i:a_i=1} c_i + \lambda \cdot \left[ P(\gamma + 1|s) \cdot c_F + \sum_{s' \in S} \sum_{o \in \mathcal{O}} P(s'|s) Q_a(o|s') \cdot V_{t+1}^*(\beta[\pi_t, a, o]) \right] \right], \quad (5)$$

$$Z(\pi_t) = c_R + \lambda \cdot V_{t+1}^*(e_0), \quad (6)$$

$$V_t^*(\pi_t) = \min \{ Y(\pi_t), W^a(\pi_t), Z(\pi_t) : a = 1, \dots, 2^D - 1 \}. \quad (7)$$

Specifically, in Eq. (4), under action  $a = 0$  (do nothing), no immediate cost is incurred. If the system fails, with probability  $P(\gamma + 1|s)$  starting from state  $s$ , the silent failure cost of  $c_F$  incurs at the end of the period. Finally, the system transitions to a new core state  $s' \in S$ , the belief is updated per Eq. (3), and the total expected discounted cost-to-go  $V_{t+1}^*(\beta[\pi_t, 0, -1])$  incurs starting from the updated belief. In Eq. (5), under action  $a \in \{1, \dots, 2^D - 1\}$  (monitor the system), the immediate data observation cost of  $\sum_{i:a_i=1} c_i$  incurs because of using the selected subset of sensors to monitor the system. Similar to Eq. (4), if the system fails, the silent failure cost of  $c_F$  incurs at the end of the period. Finally, the system transitions to a new core state  $s' \in S$  and observed system state  $o$  is made, based on which the belief is updated



**Fig. 1.** PBVI algorithm to solve the proposed POMDP model.

per Eq. (3), and the total expected discounted cost-to-go  $V_{t+1}^*(\beta|\pi_t, a, o)$  incurs starting from the updated belief. In Eq. (6), under action  $a = 2^D$  (system replacement), the system is replaced at the one-time cost of  $c_R$  and the process continues with a brand new system. That is, the system starts from belief  $e_0$  in the next period and incurs the total expected discounted cost-to-go of  $V_{t+1}^*(e_0)$ . Finally, Eq. (7) calculates the minimum total expected discounted cost-to-go at time  $t$  starting from belief  $\pi_t$ .

The existence of a solution for our problem is ensured by incorporating a discount factor ( $\lambda$ ) in the formulation. The presence of the discount factor  $\lambda < 1$  guarantees a well-defined finite cost in the infinite horizon. By discounting future costs, our problem avoids infinite accumulations of costs and enables the existence of an optimal solution. This property is well-documented in the literature on Markov Decision Processes and their solution methods. In particular, we draw upon the work of Puterman (2014) to support the existence of a solution in our problem domain. Puterman's comprehensive text discusses the theoretical foundations of MDPs and provides insights into the existence and computation of optimal solutions for such problems.

### 3.7. Heuristic policy

We consider two heuristic policy types: heuristic policy with pre-specified threshold and heuristic policy with pre-specified replacement

time. For heuristic policies with pre-specified threshold, the system is replaced when the probability of system failure meets or exceeds a certain priori threshold, denoted by  $\rho$ . In this heuristic policy, various configurations are possible. For instance, in a system equipped with two sensors, four basic setups can be explored: no sensor data collection (i.e., never collecting data), data collection from a pre-defined single sensor (i.e., sensor 1 or sensor 2), or simultaneous data collection from both sensors. For heuristic policy with pre-specified replacement time, the system is replaced at a pre-specified time without data collection. We use the Mean Time To Failure (MTTF) metric to specify the time for replacement. This is a commonly used metric in maintenance to estimate the average time between failures of a system or component. MTTF is often applied to repairable products or equipment, where they can be fixed and returned to service after a failure (García & Salgado, 2022).

### 3.8. Extension to multiple independent machines

The proposed model can be extended to systems with multiple "independent" machines. This extension is tailored to systems where machines are assumed to be independent of each other. In such scenarios, the condition of one machine does not influence the condition of another, simplifying the modeling process while maintaining relevance to practical situations.

To address systems with multiple independent machines, we propose running the optimization process separately for each machine. This approach results in multiple instances of our existing model, each customized to optimize the selection of sensors for monitoring a specific machine's condition. This adaptation allows us to consider the heterogeneous nature of sensors and the distinct operational characteristics of each machine. Next, if different machines require different subsets of sensors to be fused, we can incorporate a second step into the decision-making process. After optimizing actions for each machine individually, we can introduce a fusion step where we select the optimal actions (sensor subsets) for all machines collectively. This can be based on a new criterion, such as maximizing the information gained for all systems, ensuring efficient data utilization.

#### 4. Computational results

In this section, we first derive dynamic sensor selection and maintenance policies for a small system with two sensors and conduct extensive sensitivity analysis to draw insights. We next conduct expanded numerical experiments for a system with ten sensors and compare the results with heuristic policies.

To solve the POMDP models, we apply point based value iteration (PBVI) method (Pineau, Gordon, & Thrun, et al., 2003). PBVI is an approximate solution method that iteratively selects and expands a set of representative belief points, to which it applies value iteration updates to obtain the best corresponding total expected costs and actions (Pineau et al., 2003). Fig. 1 presents the pseudo-code for the PBVI algorithm (Pineau, Gordon, & Thrun, 2005) to solve the proposed POMDP model. All numerical analyses are implemented in Python 3.7.

Note that the first step in backup operation creates  $|A||O||\Gamma_{t-1}|$  projections (in time  $|S|^2|A||O||\Gamma_{t-1}|$ ). This is then reduced to at most  $|\bar{\pi}|$  (the size of the belief point set) components in later steps in the backup operation (in time  $|S||A||\Gamma_{t-1}||O||\bar{\pi}|$ ). Thus, a full point-based value update takes only polynomial time. Also, note that belief expansion (step 4 in Fig. 1) is much less expensive than backup operation. This makes PBVI an attractive solution method for large scale POMDP models (Pineau, Gordon, & Thrun, 2006).

##### 4.1. System with two sensors

In this section, we introduce a system that is monitored by two sensors. Specifically, in Section 4.1.1, we compare the optimal policy with a series of straightforward heuristic policies to provide insights. In Section 4.1.2, we provide a pictorial representation of the structure of the optimal policy under the entire action space,  $A$ , and a restricted action space where the number of sensors to be used is limited to 1, i.e.,  $A_1^R$ , and conduct thorough sensitivity analysis.

##### 4.1.1. An illustrative example and comparison with heuristic policies

Consider a system with two sensors. The system is subject to degradation, with the three states of 'good', 'bad', and 'failed', corresponding to '0', '1', '2'. For the purpose of the numerical experiments, the system mean in states '0', '1', and '2', are arbitrarily set to 5, 6, and 7, respectively. In the assumed model, the system degradation occurs according to the TPM  $P = \begin{pmatrix} 0.8 & 0.19 & 0.01 \\ 0 & 0.95 & 0.05 \\ 0 & 0 & 1 \end{pmatrix}$ . Values in the TPM are chosen for illustration in numerical experiments. In practice, transition probabilities are determined based on a deep understanding of the system, often estimated through data analysis, expert knowledge, or other accurate methods.

It is assumed that the system failure is silent and the state of system can only be partially observed using its two sensors. Sensor 1 is generally more accurate in good and bad states and performs relatively poorly when the system is in the failed state, i.e.,  $\sigma_1^2 = [1, 1, 20]$ . Alternatively, sensor 2 is generally more accurate in bad and failed states and performs relatively poorly when the system is in the good state, i.e.,  $\sigma_2^2 = [20, 1, 1]$ . The sample size for each sensor is set at  $n = 20$ .

Every period that the system is in the failed state, it incurs  $c_F = 1$ . At every time epoch, data can be processed from sensors at a minimal cost,  $c_1 = c_2 = 0.1$ , to partially inform the state of the system. The system can be replaced at a cost of  $c_R = 5$ . The discount factor is set to  $\lambda = 0.97$ .

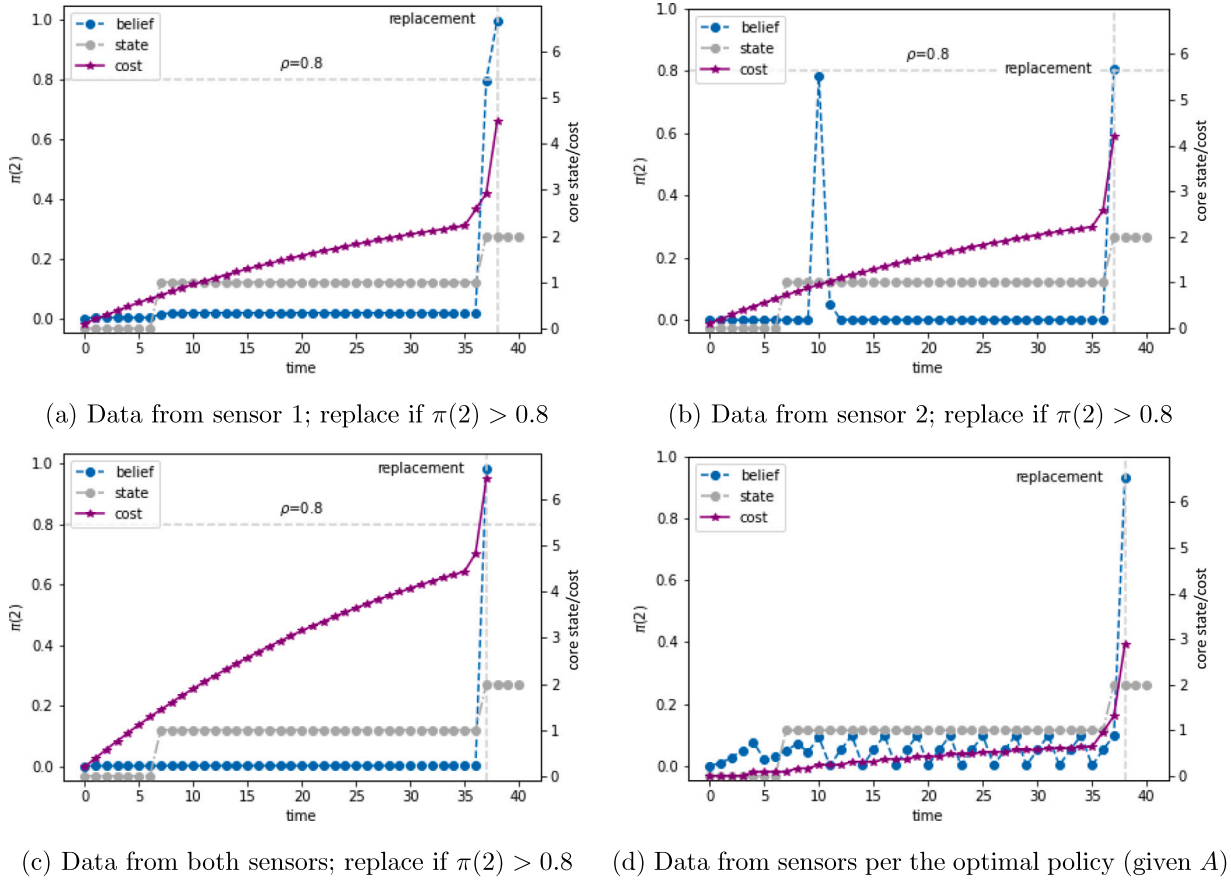
First, we solve the proposed POMDP problem to (near-)optimality using PBVI algorithm under two action sets: (1) the entire action space,  $A$  (i.e., all possible combinations of sensors (or no sensor) can be used for monitoring), and (2) a restricted action space,  $A_1^R$ , where only up to one sensor in a given epoch can be used for monitoring. We then compare the results with those under heuristic policies. For each heuristic policy, we use Eq. (3) (per the action taken) to update the system state belief. We assume the system starts in 'good' state at the beginning of decision horizon, i.e.,  $\pi = [1, 0, 0]$ . We simulate 1000 sample paths for a total number of 1000 time epochs,  $T = 1000$ , at which point the net present value of a unit of cost is much smaller than  $10^{-10}$ . Finally, we report the average and standard deviation of the total expected discounted costs. To calculate the MTTF, we conduct simulations of the system 1000 times and obtained the average MTTF from these runs.

Table 1 presents total expected discounted costs for the optimal policy under both the full action space  $A$  and the restricted action space  $A_1^R$ , alongside five heuristic policies. As seen in this table, the optimal policy significantly outperforms the heuristic policies with pre-specified thresholds ( $t$ -test  $p$ -value  $< 10^{-5}$ ) across nine  $\rho$  values from 0.1 to 0.9. The results are only presented for  $\rho = 0.6, 0.7, 0.8$ , and 0.9. The analysis confirms that lower  $\rho$  values generally yield higher expected costs or statistically insignificant differences ( $t$ -test  $p$ -value  $> 0.01$ ). As expected, collecting data is generally more beneficial than never collecting data. However, collecting data from both sensors is very costly; the reduction of silent failure cost that is achieved by more accurate observation of the system state is traded off by the total cost of observations. Recall that sensor 2 is more accurate than sensor 1 in the failed state. Therefore, collecting data only from sensor 2 is expected to be the best heuristic policy as sensor 2 can more accurately infer the system silent failure, resulting in a more timely system replacement. This is consistent with the results in Table 1. The results of the MTTF suggests that, although it outperforms the fixed threshold heuristic policies in some cases (e.g.  $13.39 \pm 5.85$  under the MTTF policy vs.  $14.28 \pm 7.30$  under the never collect data with  $\rho = 0.8$ ), it underperforms the POMDP model. Lastly, it is interesting to note that the results under the optimal policy, given the entire action space,  $A$ , and the restricted action space,  $A_1^R$ , are comparable ( $t$ -test  $p$ -value = 0.74).

Fig. 2 provides example sample paths, representing the system degradation, and the corresponding belief of failure and costs under three heuristic policies and the optimal policy (given  $A$ ). Specifically, in Fig. 2(a), the heuristic policy of always using sensor 1 maintains consistently low belief in failure when the system is in bad and good states. However, when transitioning to the failed state at time epoch 37, the belief ( $\pi(2) = 0.7953$ ) does not trigger timely replacement ( $\pi(2) \leq 0.8$ ). Only after another observation does the updated belief exceed the  $\rho = 0.8$  threshold ( $\pi(2) = 0.9956$ ). This delay in detecting system failure is mainly due to the relatively lower accuracy of sensor 1 in the failed state ( $\sigma_1^2(2) = 20$ ), resulting in incurring the silent failure cost twice, totaling 4.49 in discounted cost.

In Fig. 2(b), the heuristic policy of always using sensor 2 maintains consistently low belief in failure during bad and good states. There is a temporary increase in belief at time epoch 10 due to an erroneous observation ( $\pi(2) = 0.7822$ ), promptly corrected at time epoch 11. When the system transitions to the failed state at time epoch 37, the belief in the failed state sharply rises ( $\pi(2) = 0.8061$ ), surpassing  $\rho = 0.8$ . This is due to sensor 2's high accuracy in the failed state ( $\sigma_2^2(2) = 1$ ). Consequently, the silent failure cost is incurred for only one time epoch, resulting in a slightly lower total expected discounted cost of 4.20, compared to using data from sensor 1.

Under the heuristic policy that prescribes to always use both sensors, which is presented in Fig. 2(c), the belief of failure is overall



**Fig. 2.** Sample path of the system states and the corresponding beliefs of failure and costs under various data collection schemes. The left axis shows the belief of failure. The right axis shows the core state and total expected discounted cost. The values on the plots show the system states and failure belief at the beginning of period  $t$  and the associated cost incurred in period  $[t, t+1)$ .

**Table 1**

Total expected discounted cost ( $\pm$  standard deviation) under the optimal policy (obtained using the PBVI algorithm) for a system with two sensors, given the entire action space,  $A$ , and the restricted action space,  $A_1^{\mathcal{R}}$ , and heuristic policies.

|                                                                      | Total expected discounted cost |                  |                  |                  |
|----------------------------------------------------------------------|--------------------------------|------------------|------------------|------------------|
| Heuristic policy with pre-specified threshold for system replacement | $\rho = 0.6$                   | $\rho = 0.7$     | $\rho = 0.8$     | $\rho = 0.9$     |
| Never collect data                                                   | $13.11 \pm 5.52$               | $13.88 \pm 6.16$ | $14.28 \pm 7.03$ | $15.74 \pm 7.38$ |
| Collect data from sensor 1                                           | $11.14 \pm 3.60$               | $11.01 \pm 3.48$ | $11.03 \pm 3.83$ | $11.24 \pm 4.05$ |
| Collect data from sensor 2                                           | $10.93 \pm 3.39$               | $11.2 \pm 3.61$  | $10.77 \pm 3.36$ | $10.75 \pm 3.63$ |
| Collect data from both sensors                                       | $13.21 \pm 3.15$               | $13.37 \pm 3.45$ | $13.21 \pm 3.39$ | $13.24 \pm 3.33$ |
| Heuristic policy with pre-specified replacement time                 |                                |                  |                  |                  |
| MTTF                                                                 | $13.39 \pm 5.85$               |                  |                  |                  |
| Optimal policy                                                       |                                |                  |                  |                  |
| Under $A$                                                            | $8.93 \pm 3.39$                |                  |                  |                  |
| Under $A_1^{\mathcal{R}}$                                            | $8.87 \pm 3.82$                |                  |                  |                  |

more accurate, compared with those in Fig. 2(a)–(b). For instance, the temporary increase in belief of failure as a result of an erroneous observation by sensor 2 in time epoch 10 seen in Fig. 2(b) is now nonexistent. Similarly, the delayed increase in the belief of system failure seen in Fig. 2(a) is now fixed; the belief of failure now sharply increases to  $\pi(2) = 0.9853$  following system failure at time epoch 37. This increased accuracy is, however, gained at the additional cost of using both sensors to monitor the system. Indeed, the total expected discounted cost is now higher (at 6.45) than those incurred when monitoring the system using only sensor 1 (at 4.49) or sensor 2 (at 4.20).

Fig. 2(d) illustrates the sample path under the optimal policy (with the entire action space,  $A$ ). While the belief of system failure remains generally low in the good and bad states, it is not as close to zero as in Fig. 2(a)–(c). This is because under the optimal policy, the system is not

continuously observed; rather, in this example, the system is observed on average every 3–5 epochs, in the first 10 time epochs with sensor 1 and afterwards with sensor 2. Additionally, on average, the frequency of collecting data increases as time goes by, with a sampling frequency of every 5 epochs at the beginning to the increased frequency of every 3 epochs toward the end. As a result, the belief of being in the failed state is accurately adjusted to be closer to zero, e.g., in time epochs 5, 9, 11, 14, etc. Finally, as the system transitions to the failed state at time epoch 37, the failure goes undetected for one time epoch as no data are collected. However, as the belief of failure rises, the optimal policy prescribes data collection (with sensor 2 alone) at time epoch 38, at which point the belief of being in the failed state sharply rises ( $\pi(2) = 0.9294$ ) and a replacement is prescribed. Under the optimal policy, the cost of silent failure is slightly larger compared with when data are always collected from either sensor 2 or both sensors (Fig. 2(b)–(c),



**Table 2**

Parameter values for the problem settings used in the sensitivity analysis. All cases use Baseline parameter values, unless specified otherwise.

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baseline  | $n = 20, \lambda = 0.97, c_F = 1, c_R = 5, c_1 = c_2 = 0.1, \sigma_1^2 = [1, 1, 20], \sigma_2^2 = [20, 1, 1], \mu = [5, 6, 7],$<br>$P = \begin{pmatrix} 0.8 & 0.19 & 0.01 \\ 0 & 0.95 & 0.05 \\ 0 & 0 & 1 \end{pmatrix}, Q_1 = \begin{pmatrix} 0.99 & 0.01 & 0 \\ 0.013 & 0.974 & 0.013 \\ 0.07 & 0.24 & 0.69 \end{pmatrix}, Q_2 = \begin{pmatrix} 0.69 & 0.24 & 0.07 \\ 0.013 & 0.974 & 0.013 \\ 0 & 0.01 & 0.99 \end{pmatrix}, Q_3 = \begin{pmatrix} 0.99 & 0.01 & 0 \\ 0 & 1 & 0 \\ 0 & 0.01 & 0.99 \end{pmatrix}$                                          |
| Case I    | Faster deterioration (stochastically larger TPM); $P = \begin{pmatrix} 0.7 & 0.25 & 0.05 \\ 0 & 0.9 & 0.1 \\ 0 & 0 & 1 \end{pmatrix}$                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Case II   | Higher replacement cost; $c_R = 10$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Case III  | Lower sensor variance; $\sigma_1^2 = [1, 1, 10], \sigma_2^2 = [10, 1, 1]$<br>$Q_1 = \begin{pmatrix} 0.99 & 0.01 & 0 \\ 0.013 & 0.974 & 0.013 \\ 0.02 & 0.22 & 0.76 \end{pmatrix}, Q_2 = \begin{pmatrix} 0.76 & 0.22 & 0.02 \\ 0.013 & 0.974 & 0.013 \\ 0 & 0.01 & 0.99 \end{pmatrix}, Q_3 = \begin{pmatrix} 0.99 & 0.01 & 0 \\ 0 & 1 & 0 \\ 0 & 0.01 & 0.99 \end{pmatrix}$                                                                                                                                                                                     |
| Case IV   | Higher per-period data observation cost; $c_1 = c_2 = 0.25$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Case V    | Higher sensor variance in the 'bad' state; $\sigma_1^2 = [1, 10, 20], \sigma_2^2 = [20, 10, 1]$<br>$Q_1 = \begin{pmatrix} 0.99 & 0.01 & 0 \\ 0.24 & 0.52 & 0.24 \\ 0.07 & 0.24 & 0.69 \end{pmatrix}, Q_2 = \begin{pmatrix} 0.69 & 0.24 & 0.07 \\ 0.24 & 0.52 & 0.24 \\ 0 & 0.01 & 0.99 \end{pmatrix}, Q_3 = \begin{pmatrix} 0.99 & 0.01 & 0 \\ 0.16 & 0.68 & 0.16 \\ 0 & 0.01 & 0.99 \end{pmatrix}$                                                                                                                                                            |
| Case VI   | Possibility of system spontaneous repair in the 'bad' state; $P = \begin{pmatrix} 0.8 & 0.19 & 0.01 \\ 0.025 & 0.925 & 0.05 \\ 0 & 0 & 1 \end{pmatrix}$                                                                                                                                                                                                                                                                                                                                                                                                        |
| Case VII  | $c_1 = 0.1, c_2 = 0.05$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Case VIII | $c_1 = 0.05, c_2 = 0.1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Case IX   | Faster deterioration; $P = \begin{pmatrix} 0.7 & 0.25 & 0.05 \\ 0 & 0.9 & 0.1 \\ 0 & 0 & 1 \end{pmatrix}$ , lower sensor 2 cost; $c_2 = 0.05$                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Case X    | Faster deterioration; $P = \begin{pmatrix} 0.7 & 0.25 & 0.05 \\ 0 & 0.9 & 0.1 \\ 0 & 0 & 1 \end{pmatrix}$ , lower sensor 1 cost; $c_1 = 0.05$                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Case XI   | Higher sensor variance in the 'bad' state; $\sigma_1^2 = [1, 10, 20], \sigma_2^2 = [20, 10, 1]$<br>$Q_1 = \begin{pmatrix} 0.99 & 0.01 & 0 \\ 0.24 & 0.52 & 0.24 \\ 0.07 & 0.24 & 0.69 \end{pmatrix}, Q_2 = \begin{pmatrix} 0.69 & 0.24 & 0.07 \\ 0.24 & 0.52 & 0.24 \\ 0 & 0.01 & 0.99 \end{pmatrix}, Q_3 = \begin{pmatrix} 0.99 & 0.01 & 0 \\ 0.16 & 0.68 & 0.16 \\ 0 & 0.01 & 0.99 \end{pmatrix}$<br>Possibility of system spontaneous repair in the 'bad' state; $P = \begin{pmatrix} 0.8 & 0.19 & 0.01 \\ 0.025 & 0.925 & 0.05 \\ 0 & 0 & 1 \end{pmatrix}$ |

respectively). However, the cost of observation is substantially lower, which results in an overall lowest cost (at 2.88) compared with those obtained from the policies examined (ranging from 4.20–6.45).

#### 4.1.2. Sensitivity analysis

Here, we conduct extensive sensitivity analyses and provide insights into the impact of parameter changes on the optimal policy. Table 2 provides the parameter values under the case introduced earlier in Section 4.1.1, which we refer to as Baseline in the remainder of this section. In addition, it provides additional parameter settings, which we refer to as Cases I–XI. In Cases I–VIII, we conduct separate evaluations to assess the individual impact of important parameters. However, in Cases IX–XI, we adopt a comprehensive approach by simultaneously evaluating the impact of multiple parameters. This allows us to capture the interaction effects among these parameters and examine their influence on the optimal policy. In the following, we discuss the structure of the optimal policy under Baseline and elaborate on the impact of parameter changes on this observed structure.

Fig. 3 provides a pictorial representation of the optimal policy lookup table in which each belief state is mapped to a prescribed action. Specifically, it provides the optimal policy for Baseline given both the entire action space  $A$  and the restricted action space  $A_{\mathcal{R}}$ , as well as the optimal policy for Cases I–XI, introduced in Table 2, under the entire action space  $A$ . In this figure, belief of being in the bad state  $\pi(1)$  is plotted against the belief of being in the good state  $\pi(0)$ . Recall that  $\pi(0) + \pi(1) + \pi(2) = 1$ . Hence, the belief of being in the failed state,  $\pi(2)$ , can be easily inferred.

As seen in Fig. 3(a), when the belief of system failure is fairly low (i.e.,  $\pi(0) + \pi(1) \approx 1$ ), it is optimal to forego data collection entirely (i.e., action 0 is optimal). When the system is generally believed to be in the good state (relatively large  $\pi(0)$ ), it is optimal to use sensor 1 only (i.e., action 1 is optimal). This is because sensor 1 is more accurate than sensor 2 when the system is in the good state ( $\sigma_1^2(0) = 1 \ll \sigma_2^2(0) = 20$ ). Although using both sensors provide s additional accuracy, the minimal added accuracy gained by using sensor 2 is not necessarily worth the additional cost incurred. As the uncertainty in the belief of being in the three states (i.e., belief entropy) increases, it becomes optimal to collect data from both sensors. (i.e., action 3 is optimal). Next, when the belief of being in the good state is relatively low ( $\pi(0) < 0.35$ ), with the belief of being in the bad state not being too high or too low ( $0.15 <$

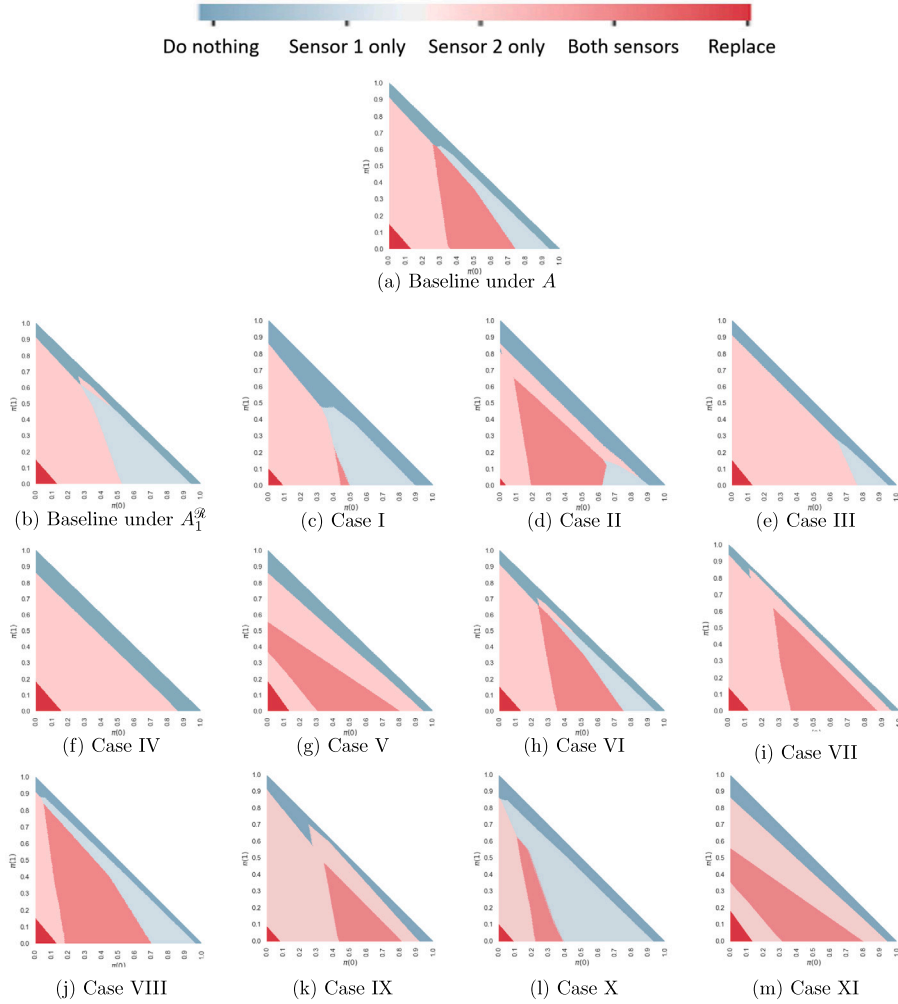
$\pi(1) < 0.9$ ), the optimal policy prescribes collecting data from sensor 2 only. Note that when the system is in the bad state, it is mathematically equivalent to collect data from sensor 1 or sensor 2 as  $\sigma_1^2(1) = \sigma_2^2(1) = 1$ . Without loss of generality, in our experiments, we prioritize sensors with a larger index to break ties. In addition, in the failed state sensor 2 is more accurate than sensor 1 ( $\sigma_1^2(2) = 20 \gg \sigma_2^2(2) = 1$ ). Therefore, in these cases, it is optimal to use sensor 2 only as the marginal benefit of using both sensors does not justify the additional data observation cost. Finally, when the probability of being in the failed state is relatively large (i.e.,  $\pi(0) + \pi(1) < 0.13$ ), the optimal policy prescribes a system replacement.

Fig. 3(b) plots the optimal policy under Baseline, but in this case, the action space is restricted to only allow for using at most one sensor for monitoring, i.e., both sensors cannot be used simultaneously. Hence, as seen in the figure, in the states where the system used to be monitored by both sensors in Fig. 3(a), it is now only monitored by one of the two sensors in Fig. 3(b). Specifically, for these states in Fig. 3(b), sensor 1 is mostly used when the likelihood of being in the good state is comparatively high; in contrast, sensor 2 is mostly used when the likelihood of being in the good state is comparatively low.

Fig. 3(c) presents the optimal policy where the system is subject to faster deterioration, i.e., the TPM is stochastically larger. The TPM  $P$  is said to be stochastically larger than the TPM  $P'$  if  $\sum_{j=k}^N P'_{i,j} \leq \sum_{j=k}^N P_{i,j}$  for any  $i$  and  $k$  in  $N$  (the number of core state). As seen in the figure, compared with Fig. 3(a), the optimal policy is generally more selective with respect to prescribing data collection and replacement actions, where the replacement action is only reserved for relatively high belief of failure. This is mainly because there is overall less uncertainty about the state of this fast deteriorating system as it more quickly transitions to the absorbing state of failure, hence, generally there is more limited need for data collection.

Fig. 3(d) plots the optimal policy where the system is subject to higher replacement cost. As seen in the figure, compared with Fig. 3(a), the optimal policy generally prescribes a more aggressive data collection scheme as the system stochastically degrades to more accurately learn/estimate the likelihood of system failure. However, it reserves the replacement action only for when the belief of failure is very high as such action is now more expensive.

Fig. 3(e) plots the optimal policy where the system is subject to lower sensor variance. As seen in the figure, compared with Fig. 3(a),



**Fig. 3.** Optimal policy for baseline, under both action spaces  $A$  and  $A_1^Q$ , and Cases I through XI, under action space  $A$ . Baseline and Cases I-XI are introduced in Table 2. Note that in all cases, when the probability of system failure,  $\pi(2) = 1 - (\pi(0) + \pi(1))$ , is relatively high, the recommended action is to replace the system. On the other hand, when the probability of system failure is relatively low (i.e., the diagonal or when  $\pi(0) + \pi(1) \approx 1$ ), it is optimal to forgo data collection. Lastly, in other states in the belief space, the optimal action is to use certain combinations of sensors (depending on the sensor variances and costs) to collect data.

the optimal policy generally does not prescribe pooling the sensors anymore as now using one sensor alone provides a more accurate estimate of the system state.

Fig. 3(f) plots the optimal policy where the system is subject to higher data observation cost. Similar to the previous case depicted in Fig. 3(d), the optimal policy generally does not prescribe pooling the sensors as collecting data from both sensors is expensive. However, different from Fig. 3(a) and (d), the optimal policy is overall more selective about data collection in the first place and hence, the beliefs in which action ‘0’ (no data collection) is optimal are expanded. Additionally, sensor 2 is selected for data collection as this sensor is prioritized in the bad state and is more accurate in the failed state, where more careful inference is critical.

Fig. 3(g) plots the optimal policy where the system is subject to higher variance in the ‘bad’ state. As seen in the figure, compared with Fig. 3(a), the optimal policy generally prescribes pooling sensor data from both sensors more aggressively when the belief entropy is high. This is because additional observations are required to compensate for the higher variance of each sensor in the bad state.

Fig. 3(h) plots the optimal policy where the system can stochastically repair itself in the ‘bad’ state (instead of only degrading). This

limited stochastic improvement may occur due to exogenous processes such as environmental changes, etc. As seen in the figure, compared with Fig. 3(a), the optimal policy is slightly more aggressive in terms of data collection. This is because the potential stochastic improvement in the system introduces more uncertainty to the process.

Fig. 3(i) plots the optimal policy where data observation from sensor 2 is cheaper than that in Baseline ( $c_2 = 0.05$ ). Hence, as expected, in general the optimal policy recommends to more aggressively collect data, especially from sensor 2. Moreover, the optimal policy never recommends to collect data from sensor 1 alone. Rather, it generally recommends to use the cheaper sensor 2 to collect data in the states in which it is optimal to collect data from sensor 1 in Baseline.

In contrast to Fig. 3(i), (j) plots the optimal policy where data observation from sensor 1 is cheaper than that in Baseline ( $c_1 = 0.05$ ). Analogously, in this case, the optimal policy is generally more aggressive in terms of data collection, especially from the cheaper sensor 1. This again translates into more instances of data collection from sensor 1 alone and both sensors, where the model used to recommend to either not collect data at all or collect data from sensor 2 alone (in Baseline). In our computational analysis, we observed that this trend also continues as sensor 2 becomes less accurate. That is, if sensor 2 is

not only more expensive, but also less accurate, it continues to become an even less favorable option compared with sensor 1, and is less often recommended in the optimal policy.

Fig. 3(k) presents the optimal policy where the system is subject to faster deterioration, and the data observation from sensor 2 is cheaper than that in Baseline. This case combines the characteristics of Cases I and VII. As seen in this figure, the space in which both sensors are used for observation is decreased compared to Case VII, where only the cost difference is considered. This reduction occurs because the faster deterioration leads to less uncertainty about the system state, resulting in a decreased need for data collection from both sensors. As a result, the optimal policy tends to prioritize data collection from the cheaper sensor (sensor 2) and relies less on the combination of both sensors.

Fig. 3(l) illustrates the optimal policy under conditions of accelerated system deterioration, where data observation from sensor 1 is more cost-effective than in the baseline scenario. This case encompasses the key aspects observed in Cases I and VIII, further emphasizing the findings from Case IX. As depicted in the figure, sensor 1 is utilized more frequently compared to sensor 2 due to its lower cost. Additionally, similar to Case IX, the observation space for both sensors is reduced in comparison to Case VIII due to the accelerated deterioration process. This outcome provides further insights into the interplay between sensor cost, system degradation rate, and the decision-making process in the context of condition-based maintenance.

Fig. 3(m) illustrates the optimal policy for a scenario combining higher variance in the bad state (Case V) and the system's ability to stochastically repair itself in this state (Case VI). Notably, this combined effect does not substantially change the optimal policy compared to Case V in isolation. This suggests that the influence of higher variance in the bad state (Case V) outweighs the impact of the system's stochastic repair capability (Case VI) when determining the optimal policy. In essence, higher variance in the bad state exerts a more significant influence on the optimal policy than the stochastic repair capability.

We further examine the cost breakdown associated with the optimal policies for both the action space  $A$  and  $A_1^{\mathcal{R}}$ . Specifically, we analyze the data observation cost, silent failure cost, and replacement cost of the optimal policies across all cases. Overall, the findings indicate that the data observation, silent failure, and replacement costs exhibit similar statistical characteristics ( $p$ -value  $> 0.05$ ), with a few exceptions observed in the baseline, IV, V, and XI cases. For additional insights on this observation, please refer to Appendix B.

#### 4.2. System with ten sensors

In this section, we conduct expanded numerical experiments for a more practical setting with  $D = 10$  candidate sensors for monitoring the Baseline system introduced in Section 4.1. Specifically, we introduce a set of 27 candidate sensors, from which we bootstrap ten sensors at a time to generate ten cases colorblue and present the results for these ten cases. The 27 candidate sensors considered differ in terms of variance in different system states and the corresponding data observation cost. The cost of using a sensor is defined to be inversely proportional to its variances in different states, making more accurate sensors more costly. Specifically, the cost of sensor  $i$  is defined as follows:

$$\hat{c}_i = \frac{1}{\sigma_i^2(0)} + \frac{2}{\sigma_i^2(1)} + \frac{4}{\sigma_i^2(2)}, \quad i = 1, \dots, 27. \quad (8)$$

The values of  $\hat{c}_i$  are then normalized in the interval  $(0, 0.5]$ . Specifically, we divide  $\hat{c}_i$  values by  $2 \cdot \max\{\hat{c}_i\}$  to maintain the most expensive per period data observation cost by a sensor as half of the per period silent failure cost,  $c_F$ , which is set to one unit of cost ( $c_F = 1$ ). To assess sensor quality, we define it as the weighted average of the inverse variances in good, bad, and failed states (i.e.,  $\frac{1}{3}(\frac{1}{\sigma_i^2(0)} + \frac{2}{\sigma_i^2(1)} + \frac{4}{\sigma_i^2(2)})$ ). Table 3 presents the 27 candidate sensors, their quality rankings, qualities, data observation costs, and variances in different system states.

##### 4.2.1. Cases with ten randomly bootstrapped sensors

We conduct numerical experiments for ten cases (namely, Cases 1–10) using random subsets of ten sensors, selected from the 27 candidate sensors. introduced in Table 3. The optimal data fusion and maintenance policy is derived for each case given the entire action space,  $A$ , as well as restricted action spaces  $A_2^{\mathcal{R}}$  and  $A_1^{\mathcal{R}}$ . Note that restricting the number of sensors to be used at a time to two and one reduces the action space size from 1025 (i.e., full action space size) to 57 and 12, respectively (i.e.,  $|A| = 1025$  and  $|A_2^{\mathcal{R}}| = 57$ ,  $|A_1^{\mathcal{R}}| = 12$ ).

Table 4 presents the ten cases with their associated list of candidate sensors and their corresponding quality mean. Table 5 presents the results for these ten cases with full and restricted action spaces. This table includes the total number of sensors, among the ten candidate sensors, that are dynamically selected for data collection in any given case. Furthermore, the table presents the optimal sensor sets for data collection, the associated optimal total expected discounted cost, and the  $p$ -values comparing the associated costs for the full and restricted action spaces under 1000 simulation runs for  $T = 1000$  time epochs. Note that in the optimal sensor sets in Table 5, some sensor ID's are presented in parenthesis. This indicates that those sensors must be fused under the optimal policy to estimate the system status in certain belief states.

As seen in Table 5, in general it is optimal to use sensors with relatively low to medium data observation costs, which have relatively larger variances, as opposed to sensors with relatively high data observation costs, which have relatively smaller variances. Often, to compensate for the higher variances of the cheaper sensors, when the use of more than one sensor is allowed, it is optimal to fuse sensors. For instance, as seen in Table 5, in Case 1, sensors 14, 15, and 17, which have average quality per Table 3, are used in a stand-alone manner as part of optimal actions. However, sensors 24 and 27, which are of the poorest quality (i.e., highest variance) per Table 3, are often fused together, or with other sensors (e.g., 14 and 15), as part of optimal actions, when it is allowed.

In Table 5, when sensor fusion is allowed (under  $A$  and  $A_2^{\mathcal{R}}$ ), sensors are selected and fused in a manner to compensate their lower accuracies, particularly in bad and failed states. For instance, in Case 1, when the belief of system failure is low, relatively cheap but inaccurate sensors (24 and 27) are used. As the belief of failure increases (e.g.,  $\pi(2) = 0.7$ ), more inexpensive sensors (14, 15, 17, 24, and 27) are fused to gather sufficient information. Notably, highly accurate yet expensive sensors are rarely or never used unless other sensors cannot provide adequate information, particularly in bad and failed states. Specifically, these results show that in cases where the quality mean of the sensors is small (implying that available sensors are overall inaccurate) and the quality standard deviation is relatively high (implying that the quality level varies across candidate sensors), it is likely that more than two sensors at a time are used to compensate for the lower quality (e.g., cases 1 and 5 with quality levels of  $(0.79 \pm 0.71)$  and  $(0.87 \pm 0.72)$  respectively). In contrast, when the quality mean of the sensors is rather large and the quality standard deviation is relatively low, at most two sensors at a time are generally used for monitoring (e.g., cases 4 and 6 with quality levels of  $(1.01 \pm 0.47)$  and  $(0.98 \pm 0.58)$ , respectively).

Compared to the full action space, policies under the restricted action space  $A_2^{\mathcal{R}}$  generally recommend similar optimal sensor sets for monitoring, with minor variations. For example, in Case 2, the policy under  $A_2^{\mathcal{R}}$  shares the same optimal sensor sets as  $A$ , except for the inclusion of sensor 18, 20, and 26. However, in cases where more than two sensors are simultaneously required under  $A$ ,  $A_2^{\mathcal{R}}$  introduces new two-sensor sets to adapt to the restriction. Under the restricted action space  $A_1^{\mathcal{R}}$ , where sensor fusion is not possible, in most cases a more accurate, but more costly, sensor is among the set of optimal sensors for system monitoring, e.g., sensor 3 in Case 3, sensor 2 in Cases 4 and 7, sensor 19 in Case 5, and sensor 22 in Case 10, to name a few.

**Table 3**

Set of candidate sensors and their associated qualities, data observation costs, and variances.

| Sensor quality ranking | Sensor quality | Data observation cost per unit time | Sensor variance in system state |     |        | Sensor ID |
|------------------------|----------------|-------------------------------------|---------------------------------|-----|--------|-----------|
|                        |                |                                     | Good                            | Bad | Failed |           |
| 1                      | 2.33333        | 0.50000                             | 1                               | 1   | 1      | 1         |
| 2                      | 2.03333        | 0.43571                             | 10                              | 1   | 1      | 10        |
| 3                      | 2.01667        | 0.43214                             | 20                              | 1   | 1      | 19        |
| 4                      | 1.73333        | 0.37143                             | 1                               | 10  | 1      | 4         |
| 5                      | 1.70000        | 0.36429                             | 1                               | 20  | 1      | 7         |
| 6                      | 1.43333        | 0.30714                             | 10                              | 10  | 1      | 13        |
| 7                      | 1.41667        | 0.30357                             | 20                              | 10  | 1      | 22        |
| 8                      | 1.40000        | 0.30000                             | 10                              | 20  | 1      | 16        |
| 9                      | 1.38333        | 0.29643                             | 20                              | 20  | 1      | 25        |
| 10                     | 1.13333        | 0.24286                             | 1                               | 1   | 10     | 2         |
| 11                     | 1.06667        | 0.22857                             | 1                               | 1   | 20     | 3         |
| 12                     | 0.83333        | 0.17857                             | 10                              | 1   | 10     | 11        |
| 13                     | 0.81667        | 0.17500                             | 20                              | 1   | 10     | 20        |
| 14                     | 0.76667        | 0.16429                             | 10                              | 1   | 20     | 12        |
| 15                     | 0.75000        | 0.16071                             | 20                              | 1   | 20     | 21        |
| 16                     | 0.53333        | 0.11429                             | 1                               | 10  | 10     | 5         |
| 17                     | 0.50000        | 0.10714                             | 1                               | 20  | 10     | 8         |
| 18                     | 0.46667        | 0.10000                             | 1                               | 10  | 20     | 6         |
| 19                     | 0.43333        | 0.09286                             | 1                               | 20  | 20     | 9         |
| 20                     | 0.23333        | 0.05000                             | 10                              | 10  | 10     | 14        |
| 21                     | 0.21667        | 0.04643                             | 20                              | 10  | 10     | 23        |
| 22                     | 0.20000        | 0.04286                             | 10                              | 20  | 10     | 17        |
| 23                     | 0.18333        | 0.03929                             | 20                              | 20  | 10     | 26        |
| 24                     | 0.16667        | 0.03571                             | 10                              | 10  | 20     | 15        |
| 25                     | 0.15000        | 0.03214                             | 20                              | 10  | 20     | 24        |
| 26                     | 0.13333        | 0.02857                             | 10                              | 20  | 20     | 18        |
| 27                     | 0.11667        | 0.02500                             | 20                              | 20  | 20     | 27        |

**Table 4**

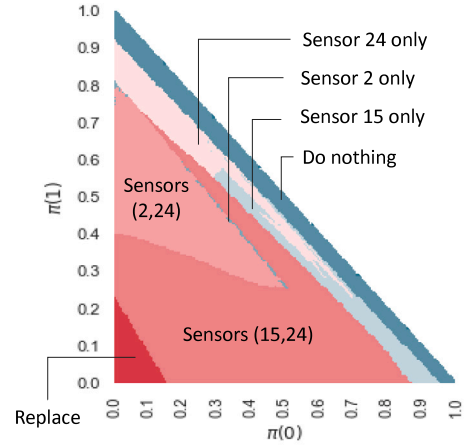
Candidate set of sensors and the associated quality mean and standard deviation. Candidate sensors are sorted in ascending order according to data observation cost.

| Case | Candidate sensors           | Quality (mean $\pm$ sd) |
|------|-----------------------------|-------------------------|
| 1    | 27,24,15,17,14,8,25,22,7,19 | 0.79 $\pm$ 0.71         |
| 2    | 18,26,9,20,25,16,7,4,19,1   | 1.21 $\pm$ 0.74         |
| 3    | 27,15,14,8,21,20,3,22,4,19  | 0.88 $\pm$ 0.63         |
| 4    | 15,9,21,20,11,2,25,16,22,4  | 1.01 $\pm$ 0.47         |
| 5    | 18,24,15,26,8,21,25,7,4,19  | 0.87 $\pm$ 0.72         |
| 6    | 15,23,6,5,3,2,25,22,13,19   | 0.98 $\pm$ 0.58         |
| 7    | 24,15,5,12,20,2,25,16,7,4   | 0.98 $\pm$ 0.55         |
| 8    | 26,23,9,12,3,25,22,13,4,19  | 1.06 $\pm$ 0.61         |
| 9    | 15,26,9,6,20,2,13,7,10,1    | 1.07 $\pm$ 0.74         |
| 10   | 24,15,9,5,2,16,22,7,4,1     | 1.10 $\pm$ 0.71         |

Overall, it is interesting to note that in the ten cases shown in Table 5, although there are instances of 3- to 5-tuple solutions under  $A$ , these solutions are not generally exploited in action. This is mainly because the belief states in which these solutions are optimal are not frequently visited. For additional insights on this observation, please refer to Appendix C.

We also investigate the cost breakdown under optimal policies for action spaces  $A$  and  $A_2^R$ . Specifically, we calculate the data observation cost, silent failure cost, and the replacement cost of the optimal policies separately for the ten cases. The results generally suggest that the data observation costs under the two action spaces are statistically different ( $p$ -value  $< 0.05$ , except for Cases 4, 7, and 10), whereas silent failure and replacement costs are generally statistically similar ( $p$ -value  $> 0.05$ , except for Cases 6 and 8 in the former and case 1 in the latter). For additional insights on this observation, please refer to Appendix D.

Fig. 4 shows the optimal policy for case 10 under action space  $A_2^R$ . Similar to the two-sensor case, when the belief of failure is low ( $\pi(2) < 0.1$ ), the optimal action is to avoid data collection. Conversely, when the belief of failure is high ( $\pi(2) > 0.8$ ), replacement is the optimal choice. For intermediate beliefs, data collection is recommended. Initially, with a relatively low probability of failure, the optimal policy utilizes a single sensor for data collection, such as sensors 2, 15, or 24. In regions where failure becomes more likely (moderate belief of

**Fig. 4.** Optimal policy under the restricted action space,  $A_2^R$ , for Case 10.

bad states and low belief of good states), sensor 24 pairs with sensor 2, which offers higher accuracy ( $\sigma_2^2 = (1, 1, 10)$ ). Conversely, in areas where the system is more likely to be in good condition, sensor 24 is paired with sensor 15, which is cost-effective but less accurate ( $\sigma_{15}^2 = (10, 10, 20)$ ).

#### 4.2.2. An illustrative case with ten cheapest sensors

To further explore possible cost savings under full action space when compared with restricted action spaces, in this section, we consider a new case in which the ten cheapest (least accurate) sensors in Table 3 are selected as candidate sensors. Our hypothesis is that because the candidate sensors in this case are relatively less accurate (compared with the ten cases used in Section 4.2.1), it is essential to fuse them to obtain an accurate estimate of the system state. Therefore, restricting the number of sensors to be used at a given epoch will adversely impact system state estimation. This will subsequently cause a significant difference in the total expected discounted costs under the full and restricted action spaces.



**Table 5**

Candidate sensors, the choice of action space (full, or restricted with up to two or one sensor), the total number of sensors, among the ten candidate sensors, that are used for data collection, optimal sensor sets for data collection, the associated total expected discounted cost ( $\pm$  standard deviation) under the optimal policies, and the  $p$ -values comparing the associated total expected discounted costs under the full and restricted action spaces for Cases 1–10. The results indicate that the associated total expected discounted cost under full and restricted action spaces are not significantly different for these ten cases.

| Case | Action space      | Number of sensors used | Optimal sensor sets for data collection                                                                     | Optimal total expected discounted cost (mean $\pm$ sd) | $p$ -value |
|------|-------------------|------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|------------|
| 1    | Full              | 5                      | 15,17,14,(27,24),(17,14),(27,15,14),<br>(27,17,14),(24,15,14),(24,17,14),(27,24,15,14),<br>(27,24,15,17,14) | 9.78 $\pm$ 4.33                                        | –          |
|      | Up to two sensors | 6                      | 27,15,17,14,19,(27,14),(27,19),<br>(24,15),(24,14),(15,14),(15,19),(17,14)                                  | 9.88 $\pm$ 4.09                                        | 0.59       |
|      | Up to one sensor  | 6                      | 24,15,17,14,8,19                                                                                            | 9.93 $\pm$ 4.14                                        | 0.44       |
| 2    | Full              | 3                      | 18,26,20,(18,26),(18,20),(26,20),(18,26,20)                                                                 | 9.7 $\pm$ 4.21                                         | –          |
|      | Up to two sensors | 3                      | 18,26,20,(18,26),(18,20),(26,20)                                                                            | 9.75 $\pm$ 4.08                                        | 0.79       |
|      | Up to one sensor  | 4                      | 18,26,9,20                                                                                                  | 9.53 $\pm$ 4.09                                        | 0.36       |
| 3    | Full              | 6                      | 27,15,14,8,20,(27,20),(14,21),(14,20),<br>(27,15,14),(27,14,20)                                             | 9.78 $\pm$ 4.06                                        | –          |
|      | Up to two sensors | 6                      | 27,15,14,8,20,(27,14),(27,20),(15,14),<br>(14,21),(14,20)                                                   | 9.7 $\pm$ 4.14                                         | 0.65       |
|      | Up to one sensor  | 7                      | 27,15,14,8,21,20,3                                                                                          | 9.87 $\pm$ 4.27                                        | 0.64       |
| 4    | Full              | 4                      | 15,9,20,11,(15,9),(15,20),(15,11)                                                                           | 9.7 $\pm$ 4.23                                         | –          |
|      | Up to two sensors | 4                      | 15,9,20,11,(15,9),(15,20),(15,11)                                                                           | 9.7 $\pm$ 4.23                                         | 1          |
|      | Up to one sensor  | 5                      | 15,9,20,11,2                                                                                                | 9.52 $\pm$ 4.09                                        | 0.34       |
| 5    | Full              | 6                      | 18,24,15,26,8,21,(18,24),(18,21),<br>(26,21),(18,24,15),(18,24,26),(18,26,21),(24,26,21),<br>(18,24,15,26)  | 9.7 $\pm$ 4.29                                         | –          |
|      | Up to two sensors | 5                      | 18,24,15,26,21,(18,26),(18,21),(24,15),<br>(24,26),(15,26),(26,21)                                          | 9.57 $\pm$ 4.09                                        | 0.51       |
|      | Up to one sensor  | 7                      | 18,24,15,26,8,21,19                                                                                         | 9.9 $\pm$ 4.18                                         | 0.3        |
| 6    | Full              | 5                      | 15,23,5,19,(15,23),(23,2)                                                                                   | 10.08 $\pm$ 4.33                                       | –          |
|      | Up to two sensors | 5                      | 15,23,5,19,(15,23),(23,2)                                                                                   | 9.79 $\pm$ 4.09                                        | 0.13       |
|      | Up to one sensor  | 5                      | 15,23,5,2,19                                                                                                | 10.02 $\pm$ 4.31                                       | 0.76       |
| 7    | Full              | 4                      | 24,15,5,20,(24,15),(24,20),(15,20),(24,15,20)                                                               | 9.66 $\pm$ 4.19                                        | –          |
|      | Up to two sensors | 4                      | 24,15,5,20,(24,15),(24,20),(15,20)                                                                          | 9.52 $\pm$ 4.21                                        | 0.43       |
|      | Up to one sensor  | 6                      | 24,15,5,12,20,2                                                                                             | 9.62 $\pm$ 4.14                                        | 0.8        |
| 8    | Full              | 4                      | 26,23,12,(26,23),(26,9),(26,12),(23,9),(26,23,12)                                                           | 9.81 $\pm$ 4.36                                        | –          |
|      | Up to two sensors | 4                      | 26,12,(26,23),(26,12),(23,9)                                                                                | 9.84 $\pm$ 4.1                                         | 0.86       |
|      | Up to one sensor  | 5                      | 26,23,9,12,3                                                                                                | 9.77 $\pm$ 3.99                                        | 0.84       |
| 9    | Full              | 3                      | 15,26,20,(15,26),(15,20),(26,20),(15,26,20)                                                                 | 9.55 $\pm$ 4.11                                        | –          |
|      | Up to two sensors | 4                      | 15,26,20,(15,26),(15,20),(26,9),(26,20)                                                                     | 9.71 $\pm$ 4.11                                        | 0.39       |
|      | Up to one sensor  | 6                      | 15,26,9,6,20,2                                                                                              | 9.78 $\pm$ 4.21                                        | 0.23       |
| 10   | Full              | 4                      | 24,15,2,(24,15),(24,2),(24,15,5),(24,15,2)                                                                  | 9.86 $\pm$ 4.27                                        | –          |
|      | Up to two sensors | 3                      | 24,15,2,(24,15),(24,2)                                                                                      | 9.58 $\pm$ 4.2                                         | 0.14       |
|      | Up to one sensor  | 5                      | 24,15,5,2,22                                                                                                | 9.88 $\pm$ 4.31                                        | 0.93       |

Table 6 presents the results for this case. As the results show, the optimal policy under the full action space  $A$  outperforms the optimal policy under the restricted action space  $A_1^R$  at the significance level of 0.05 ( $p$ -value = 0.02 < 0.05). Additionally, the optimal policy under the full action space  $A$  results in a lower total expected discounted cost compared with the optimal policy under the restricted action space  $A_2^R$ . The difference, however, is only statistically significant at the higher significance level of 0.1 ( $p$ -value = 0.07). Table 6 also confirms the insights captured from Table 5 regarding sensor fusion and its relation sensor quality. As the available sensors in Table 6 are the cheapest ones, the quality level of these sensors are quite low ( $0.23 \pm 0.11$ ). Hence, from the previous results, it is expected that more than two sensors are needed to be fused at a time.

#### 4.2.3. Comparison with heuristic policies

Next, we introduce heuristic policies that apply *a priori* conditions on model parameters (i.e., on belief of system failure,  $\pi(2)$ ), and MTTF to facilitate costs comparison with optimal policies under the full action

space. We limit these comparisons to full action space only, as the total expected discounted costs under the full and restricted action spaces are not significantly different in Cases 1–10.

Four heuristic policies are considered in our analyses. The first heuristic policy, denoted by  $\mathcal{E}_A^\rho$ , collects data in every decision epoch from the three cheapest available sensors and performs replacement when the belief of failure exceeds  $\rho$  (i.e.,  $\pi(2) \geq \rho$ ). The second heuristic policy, denoted by  $\mathcal{E}_M^\rho$ , is a time-based extension to the first heuristic policy. Specifically, the second heuristic,  $\mathcal{E}_M^\rho$ , collects data from three cheapest available sensors only *starting* from mid-way through the system MTTF and performs replacement when the belief of failure exceeds  $\rho$ . The third heuristic policy, denoted by  $\mathcal{L}_A^\rho$ , collects data in every decision epoch from the sensor with least variance and performs replacement when the belief of failure exceeds  $\rho$ . Lastly, the fourth heuristic policy, denoted by  $\mathcal{L}_M^\rho$ , is a time-based version to the third heuristic policy. Specifically,  $\mathcal{L}_M^\rho$  collects data from the sensor with the least variance only starting from mid-way through the system MTTF and performs replacement when the belief of failure exceeds  $\rho$ .

**Table 6**

Candidate sensors, the choice of action space (full, or restricted with up to two or one sensor), the total number of sensors, among the ten candidate sensors, that are used for data collection, optimal sensor sets for data collection, the associated total expected discounted cost ( $\pm$  standard deviation) under the optimal policies, and the  $p$ -values comparing the associated total expected discounted costs under full and restricted action spaces.

| Candidate sensors           | Action space      | Number of sensors used | Optimal sensor sets for data collection                                                                                                                                                        | Optimal total expected discounted cost (mean $\pm$ sd) | $p$ -value |
|-----------------------------|-------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|------------|
| 27,18,24,15,26,17,23,14,9,6 | Full              | 8                      | 18,24,17,23,14,(27,24),(27,24,14),(24,15,14),<br>(27,24,26,23),(27,24,23,14),(27,15,23,14),<br>(27,24,15,26,23),(27,24,15,23,14),(24,15,26,23,14),<br>(27,18,24,15,23,14), (27,24,15,26,23,14) | $9.8 \pm 4.31$                                         | –          |
|                             | Up to two sensors | 6                      | 27,24,15,23,14,(27,24),(24,15),<br>(27,23)(24,23),(24,14),(17,14),(23,14)                                                                                                                      | $10.16 \pm 4.41$                                       | 0.07       |
|                             | Up to one sensor  | 6                      | 27,24,15,26,23,14                                                                                                                                                                              | $10.26 \pm 4.49$                                       | 0.02       |

**Table 7**

List of sensors used and the associated total expected discounted cost ( $\pm$  standard deviation) under the heuristic policies  $\mathcal{E}_A^\rho$  for Cases 1–10.

| Case | Sensors used | Total expected discounted cost under policy $\mathcal{E}_A^\rho$ (mean $\pm$ sd) |                  |                  |                  |                  |
|------|--------------|----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|
|      |              | $\rho = 0.5$                                                                     | $\rho = 0.6$     | $\rho = 0.7$     | $\rho = 0.8$     | $\rho = 0.9$     |
| 1    | 15,24,27     | $12.00 \pm 3.69$                                                                 | $11.9 \pm 3.84$  | $11.72 \pm 3.8$  | $11.77 \pm 4.17$ | $11.72 \pm 4.28$ |
| 2    | 9,18,26      | $14.82 \pm 3.7$                                                                  | $14.54 \pm 3.81$ | $14.04 \pm 4.01$ | $14.03 \pm 4.09$ | $14.26 \pm 4.28$ |
| 3    | 14,15,27     | $12.54 \pm 3.71$                                                                 | $12.24 \pm 3.83$ | $12.29 \pm 3.81$ | $12.04 \pm 3.99$ | $12.2 \pm 4.07$  |
| 4    | 9,15,21      | $16.77 \pm 3.33$                                                                 | $16.75 \pm 3.37$ | $16.77 \pm 3.39$ | $16.75 \pm 3.33$ | $17.07 \pm 3.69$ |
| 5    | 18,24,15     | $12.15 \pm 3.78$                                                                 | $11.93 \pm 3.76$ | $11.82 \pm 3.84$ | $11.76 \pm 4.09$ | $11.83 \pm 4.17$ |
| 6    | 6,15,23      | $14.59 \pm 3.70$                                                                 | $14.26 \pm 3.77$ | $14.35 \pm 3.78$ | $14.32 \pm 3.9$  | $14.44 \pm 4.12$ |
| 7    | 5,15,24      | $14.52 \pm 3.71$                                                                 | $14.32 \pm 3.78$ | $14.23 \pm 3.7$  | $14.26 \pm 3.84$ | $14.35 \pm 4.07$ |
| 8    | 9,23,26      | $14.69 \pm 3.7$                                                                  | $14.78 \pm 3.72$ | $14.49 \pm 3.83$ | $14.27 \pm 4.02$ | $14.42 \pm 4.05$ |
| 9    | 9,15,26      | $14.5 \pm 3.66$                                                                  | $14.35 \pm 3.71$ | $14.13 \pm 3.89$ | $14.11 \pm 4.14$ | $14.11 \pm 4.1$  |
| 10   | 9,15,24      | $14.13 \pm 3.66$                                                                 | $13.91 \pm 3.77$ | $13.93 \pm 3.8$  | $13.88 \pm 3.96$ | $14.01 \pm 4.24$ |

**Table 8**

List of sensors used and the associated total expected discounted cost ( $\pm$  standard deviation) under the heuristic policies  $\mathcal{E}_M^\rho$  for Cases 1–10.

| Case | Sensors used | Total expected discounted cost under policy $\mathcal{E}_M^\rho$ (mean $\pm$ sd) |                  |                  |                  |                  |
|------|--------------|----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|
|      |              | $\rho = 0.5$                                                                     | $\rho = 0.6$     | $\rho = 0.7$     | $\rho = 0.8$     | $\rho = 0.9$     |
| 1    | 15,24,27     | $11.03 \pm 4.43$                                                                 | $10.89 \pm 4.52$ | $10.87 \pm 4.58$ | $11.0 \pm 4.88$  | $10.79 \pm 4.87$ |
| 2    | 9,18,26      | $11.85 \pm 4.17$                                                                 | $11.8 \pm 4.18$  | $11.83 \pm 4.26$ | $11.85 \pm 4.58$ | $12.11 \pm 4.67$ |
| 3    | 14,15,27     | $11.18 \pm 4.33$                                                                 | $10.9 \pm 4.49$  | $10.96 \pm 4.52$ | $11.03 \pm 4.54$ | $11.12 \pm 4.78$ |
| 4    | 9,15,21      | $12.27 \pm 3.94$                                                                 | $12.24 \pm 3.88$ | $12.25 \pm 3.85$ | $12.27 \pm 3.9$  | $12.6 \pm 3.86$  |
| 5    | 18,24,15     | $11.06 \pm 4.37$                                                                 | $10.97 \pm 4.5$  | $10.85 \pm 4.49$ | $11.04 \pm 4.85$ | $10.9 \pm 4.88$  |
| 6    | 6,15,23      | $11.75 \pm 4.19$                                                                 | $11.66 \pm 4.25$ | $11.71 \pm 4.27$ | $11.65 \pm 4.26$ | $12.01 \pm 4.53$ |
| 7    | 5,15,24      | $11.72 \pm 4.05$                                                                 | $11.75 \pm 4.3$  | $11.67 \pm 4.25$ | $11.65 \pm 4.22$ | $11.98 \pm 4.5$  |
| 8    | 9,23,26      | $11.81 \pm 4.14$                                                                 | $11.89 \pm 4.28$ | $11.81 \pm 4.24$ | $12.01 \pm 4.51$ | $12.11 \pm 4.59$ |
| 9    | 9,15,26      | $11.77 \pm 4.11$                                                                 | $11.84 \pm 4.25$ | $11.67 \pm 4.23$ | $11.82 \pm 4.57$ | $12.03 \pm 4.66$ |
| 10   | 9,15,24      | $11.82 \pm 4.23$                                                                 | $11.72 \pm 4.38$ | $11.53 \pm 4.25$ | $11.69 \pm 4.49$ | $11.89 \pm 4.72$ |

**Table 9**

The sensor used and the associated total expected discounted cost ( $\pm$  standard deviation) under the heuristic policies  $\mathcal{L}_A^\rho$  for Cases 1–10.

| Case      | Sensor used | Total expected discounted cost under policy $\mathcal{L}_A^\rho$ (mean $\pm$ sd) |                  |                  |                  |                  |
|-----------|-------------|----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|
|           |             | $\rho = 0.5$                                                                     | $\rho = 0.6$     | $\rho = 0.7$     | $\rho = 0.8$     | $\rho = 0.9$     |
| 1,3,5,6,8 | 19          | $21.72 \pm 3.33$                                                                 | $21.69 \pm 3.38$ | $21.69 \pm 3.32$ | $21.54 \pm 3.41$ | $21.41 \pm 3.57$ |
| 2,9,10    | 1           | $23.71 \pm 3.19$                                                                 | $23.75 \pm 3.22$ | $23.7 \pm 3.17$  | $23.65 \pm 3.21$ | $23.67 \pm 3.53$ |
| 4,7       | 4           | $21.82 \pm 3.4$                                                                  | $20.67 \pm 3.76$ | $20.6 \pm 3.76$  | $20.5 \pm 3.71$  | $20.67 \pm 4.08$ |

We simulate these heuristic policies for 1000 randomly generated sample paths and  $T = 1000$  time epochs for Cases 1–10, where  $\rho$  changes between 0.1 and 0.9 with 0.1 increment,  $\rho \in \{0.1, 0.2, \dots, 0.9\}$ , to facilitate a comparison with the optimal policies presented in Table 5. Tables 7–10 present the simulation results for these heuristic policies. Our statistical analysis ( $t$ -tests) shows that using  $\rho < 0.5$  results in worse performance compared with using  $\rho \geq 0.5$ , and hence, for brevity, we only present the results for  $\rho \geq 0.5$ . Tables 7 and 8 present all 10 cases separately as the three cheapest sensors vary across these cases. In contrast, Tables 9 and 10 present the results when using sensors 1, 4 and 19, which have the least variance in the candidate sensor pools in Cases 1–10.

Overall, the results suggest that the optimal policy outperforms all four heuristic policies for all values of  $\rho$  in all ten cases ( $t$ -test  $p$ -value  $< 10^{-5}$ ). As seen in the tables, the heuristic policies  $\mathcal{E}_M^\rho$  and  $\mathcal{L}_M^\rho$ ,

which start data collection only half-way through the system MTTF result in lower total expected discounted costs than their counterpart heuristic policies  $\mathcal{E}_A^\rho$  and  $\mathcal{L}_A^\rho$ , which always collect data. In addition, consistent with the general observations made about the behavior of the optimal policy that prioritizes fusing cheaper sensors, the heuristic policies that use three least accurate but cheapest sensors, i.e.,  $\mathcal{E}_A^\rho$  and  $\mathcal{E}_M^\rho$ , outperform those that use the most accurate but most expensive sensor, i.e.,  $\mathcal{L}_A^\rho$  and  $\mathcal{L}_M^\rho$ .

## 5. Conclusion and future works

In this study, we develop a framework for joint dynamic sensor selection and maintenance planning in partially observable deteriorating systems. Specifically, we consider a deteriorating system with costly but silent failures, which can be monitored using a set of heterogeneous,

**Table 10**The sensor used and the associated total expected discounted cost ( $\pm$  standard deviation) under the heuristic policies  $\mathcal{L}_M^\rho$  for Cases 1–10.

| Case      | Sensor used | Total expected discounted cost under policy $\mathcal{L}_M^\rho$ (mean $\pm$ sd) |                 |                  |                  |                  |
|-----------|-------------|----------------------------------------------------------------------------------|-----------------|------------------|------------------|------------------|
|           |             | $\rho = 0.5$                                                                     | $\rho = 0.6$    | $\rho = 0.7$     | $\rho = 0.8$     | $\rho = 0.9$     |
| 1,3,5,6,8 | 19          | 13.86 $\pm$ 3.38                                                                 | 13.78 $\pm$ 3.3 | 13.7 $\pm$ 3.29  | 13.89 $\pm$ 3.41 | 14.19 $\pm$ 3.29 |
| 2,9,10    | 1           | 14.63 $\pm$ 3.21                                                                 | 14.6 $\pm$ 3.2  | 14.6 $\pm$ 3.17  | 14.6 $\pm$ 3.17  | 15.05 $\pm$ 3.09 |
| 4,7       | 4           | 13.76 $\pm$ 3.47                                                                 | 13.9 $\pm$ 3.54 | 13.97 $\pm$ 3.52 | 14.13 $\pm$ 3.76 | 14.39 $\pm$ 3.72 |

**Table 11**

Cost breakdown of the optimal policies for the baseline and Cases I–XI under the full and restricted action space.

| Case     | Action space     | Total cost       | Data observation cost | Silent failure cost | Replacement cost |
|----------|------------------|------------------|-----------------------|---------------------|------------------|
| Baseline | Full             | 8.93 $\pm$ 3.39  | 1.01 $\pm$ 0.04       | 2.32 $\pm$ 1.11     | 5.61 $\pm$ 2.02  |
|          | Up to one sensor | 8.87 $\pm$ 3.82  | 0.92 $\pm$ 0.05       | 2.38 $\pm$ 1.49     | 5.57 $\pm$ 2.61  |
| I        | Full             | 16.04 $\pm$ 4.91 | 1.08 $\pm$ 0.02       | 4.42 $\pm$ 1.80     | 10.54 $\pm$ 3.44 |
|          | Up to one sensor | 16.04 $\pm$ 4.75 | 1.08 $\pm$ 0.02       | 4.52 $\pm$ 1.81     | 10.44 $\pm$ 3.29 |
| II       | Full             | 13.65 $\pm$ 6.67 | 0.70 $\pm$ 0.04       | 3.74 $\pm$ 2.20     | 9.21 $\pm$ 4.66  |
|          | Up to one sensor | 13.65 $\pm$ 6.69 | 0.70 $\pm$ 0.04       | 3.74 $\pm$ 2.19     | 9.21 $\pm$ 4.66  |
| III      | Full             | 8.76 $\pm$ 3.96  | 0.79 $\pm$ 0.09       | 2.44 $\pm$ 1.55     | 5.53 $\pm$ 2.69  |
|          | Up to one sensor | 8.76 $\pm$ 3.97  | 0.79 $\pm$ 0.09       | 2.44 $\pm$ 1.55     | 5.53 $\pm$ 2.69  |
| IV       | Full             | 9.78 $\pm$ 3.98  | 1.50 $\pm$ 0.14       | 2.92 $\pm$ 1.80     | 5.36 $\pm$ 2.54  |
|          | Up to one sensor | 9.70 $\pm$ 4.29  | 1.21 $\pm$ 0.10       | 3.73 $\pm$ 2.17     | 5.12 $\pm$ 2.48  |
| V        | Full             | 9.97 $\pm$ 4.18  | 0.94 $\pm$ 0.10       | 3.71 $\pm$ 2.18     | 5.32 $\pm$ 2.28  |
|          | Up to one sensor | 10.02 $\pm$ 4.45 | 0.81 $\pm$ 0.07       | 3.99 $\pm$ 2.39     | 5.23 $\pm$ 2.42  |
| VI       | Full             | 8.52 $\pm$ 3.89  | 0.91 $\pm$ 0.05       | 2.32 $\pm$ 1.59     | 5.29 $\pm$ 2.59  |
|          | Up to one sensor | 8.51 $\pm$ 3.89  | 0.91 $\pm$ 0.05       | 2.32 $\pm$ 1.59     | 5.28 $\pm$ 2.59  |
| VII      | Full             | 8.21 $\pm$ 3.82  | 0.51 $\pm$ 0.02       | 2.38 $\pm$ 1.40     | 5.32 $\pm$ 2.56  |
|          | Up to one sensor | 8.18 $\pm$ 3.78  | 0.52 $\pm$ 0.02       | 2.36 $\pm$ 1.43     | 5.30 $\pm$ 2.48  |
| VIII     | Full             | 8.59 $\pm$ 3.80  | 0.82 $\pm$ 0.09       | 2.28 $\pm$ 1.41     | 5.49 $\pm$ 2.67  |
|          | Up to one sensor | 8.59 $\pm$ 3.80  | 0.82 $\pm$ 0.09       | 2.28 $\pm$ 1.41     | 5.49 $\pm$ 2.66  |
| IX       | Full             | 15.48 $\pm$ 4.90 | 0.63 $\pm$ 0.04       | 4.20 $\pm$ 1.78     | 10.65 $\pm$ 3.20 |
|          | Up to one sensor | 15.48 $\pm$ 4.90 | 0.63 $\pm$ 0.04       | 4.20 $\pm$ 1.78     | 10.65 $\pm$ 3.30 |
| X        | Full             | 15.73 $\pm$ 4.73 | 0.88 $\pm$ 0.06       | 4.31 $\pm$ 1.67     | 10.54 $\pm$ 3.39 |
|          | Up to one sensor | 15.73 $\pm$ 4.73 | 0.88 $\pm$ 0.06       | 4.31 $\pm$ 1.67     | 10.54 $\pm$ 3.39 |
| XI       | Full             | 9.51 $\pm$ 4.47  | 0.96 $\pm$ 0.11       | 3.45 $\pm$ 2.19     | 5.10 $\pm$ 2.53  |
|          | Up to one sensor | 9.62 $\pm$ 4.30  | 0.75 $\pm$ 0.05       | 3.72 $\pm$ 2.29     | 5.16 $\pm$ 2.34  |

costly, and noisy sensors whose accuracy is a function of system state. The goal is to dynamically select a subset of sensors to estimate the system state over time and prescribe maintenance action based on the observed system state to minimize the total expected discounted cost of system operation.

Specifically, we consider a system with three ordered deteriorating states. We first conduct thorough analyses using two heterogeneous candidate sensors for system monitoring. We solve the problem to near-optimality using PBVI algorithm and discuss the efficacy of the proposed approach. We then conduct expanded numerical experiments using ten heterogeneous candidate sensors, and compare the results of the optimal policy under the full action space and the restricted action spaces. We also benchmark the results with those obtained from different heuristic policies. Our results suggest that following the optimal policy provides statistically significant improvements upon the heuristic policies.

This study offers valuable insights for practitioners seeking to optimize the management of deteriorating systems. Our framework offers three key advantages: cost efficiency, enhanced reliability, and informed decision-making through comparative analysis. By dynamically selecting the most relevant sensors and optimizing maintenance actions based on the observed system state, practitioners can effectively allocate resources and reduce unnecessary expenses associated with system

monitoring and maintenance. In addition, by enabling practitioners to estimate the system state more accurately, practitioners can improve reliability in detecting system failures and taking timely maintenance actions. Furthermore, by demonstrating our approach's superior performance over heuristic policies, practitioners can gain insights into its potential benefits and make informed decisions regarding their sensor selection and maintenance strategies.

An immediate future work includes validating the proposed framework in a case study using real-world data, and augmenting the maintenance actions to include inspection and allow for imperfect repairs, among others. In addition, when fusing sensors measuring different phenomena, an expected pre-established relationship between such phenomena (e.g., the relationship between temperature and pressure) may be leveraged to improve system inference and thereby make a more informed decision about the system maintenance. A future extension can include accounting for possible sensor failures and prescribing various preventive or reactive actions to mitigate the impact. In addition, it would be valuable to explore the possibility of reducing the off times of machines resulting from failures within the problem setting. This could be achieved by incorporating a constraint into the model that accounts for minimizing the duration of machine downtime. Finally, a future extension can extend the proposed model to a system with multiple independent machines.

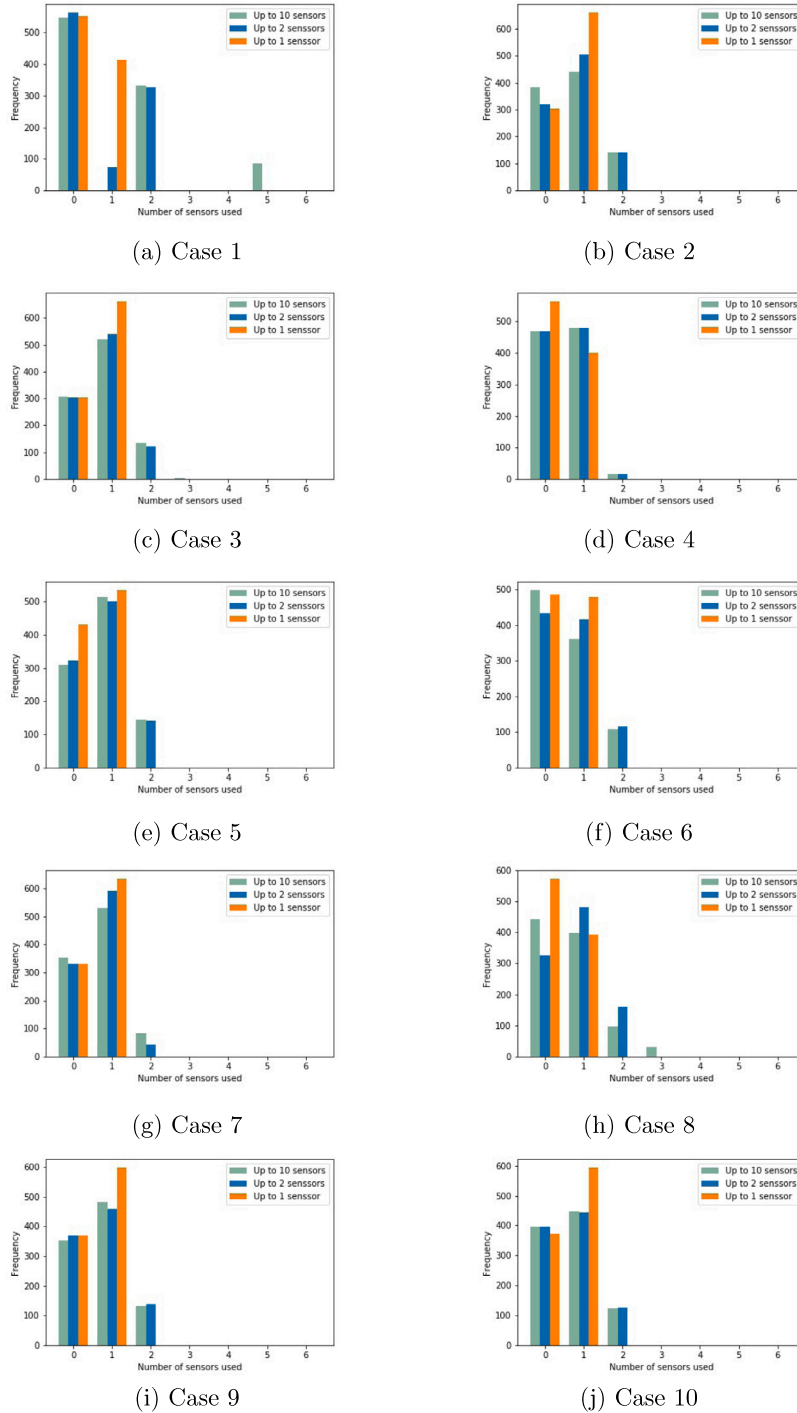


Fig. 5. Frequency of  $m$ -tuple sensor sets used for system monitoring under  $A$  (up to 10 sensors),  $A_2^R$  (up to 2 sensors), and  $A_1^R$  (up to 1 sensor) in simulation runs of Cases 1–10 with 10 candidate sensors.

#### CRedit authorship contribution statement

**Mahboubeh Madadi:** Conceptualized the study, Performed the numerical analysis and created the manuscript. **Shahrbano Rezaei:** Performed the numerical analysis and created the manuscript. **Anahita Khojandi:** Conceptualized the study, Performed the numerical analysis and created the manuscript.

#### Data availability

Data will be made available on request.

#### Appendix A. Estimating the information matrices

In this section, we propose a pre-processing (calibration) step to estimate information matrices needed in the proposed POMDP model. That is, the outputs of this pre-processing step serve as inputs to the optimization model. As discussed in Section 3, we assume sensor observation is a normal random variable whose variance in different system states are previously established through testing. When one or more sensors are used for monitoring, the system state needs to be estimated from the observed data. In this section, we present how we derive the information matrices, providing the probability of different



**Table 12**  
Cost breakdown of the optimal policies for Cases 1–10.

| Case | Action space      | Total cost   | Data observation cost | Silent failure cost | Replacement cost |
|------|-------------------|--------------|-----------------------|---------------------|------------------|
| 1    | Full              | 9.78 ± 4.33  | 1.04 ± 0.14           | 3.57 ± 2.24         | 5.18 ± 2.43      |
|      | Up to two sensors | 9.88 ± 4.09  | 1.60 ± 0.25           | 3.45 ± 1.96         | 4.84 ± 2.28      |
| 2    | Full              | 9.70 ± 4.21  | 1.30 ± 0.22           | 3.40 ± 2.08         | 5.00 ± 2.45      |
|      | Up to two sensors | 9.75 ± 4.08  | 1.39 ± 0.25           | 3.24 ± 1.98         | 5.11 ± 2.46      |
| 3    | Full              | 9.78 ± 4.06  | 1.34 ± 0.23           | 3.34 ± 2.00         | 5.10 ± 2.42      |
|      | Up to two sensors | 9.70 ± 4.14  | 1.30 ± 0.23           | 3.29 ± 2.08         | 5.11 ± 2.42      |
| 4    | Full              | 9.70 ± 4.23  | 1.26 ± 0.21           | 3.44 ± 2.19         | 5.00 ± 2.41      |
|      | Up to two sensors | 9.70 ± 4.23  | 1.26 ± 0.21           | 3.44 ± 2.19         | 5.00 ± 2.41      |
| 5    | Full              | 9.70 ± 4.29  | 1.33 ± 0.24           | 3.35 ± 2.13         | 5.03 ± 2.50      |
|      | Up to two sensors | 9.57 ± 4.09  | 1.29 ± 0.23           | 3.29 ± 1.98         | 5.00 ± 2.45      |
| 6    | Full              | 10.08 ± 4.33 | 1.40 ± 0.27           | 3.73 ± 2.23         | 4.95 ± 2.37      |
|      | Up to two sensors | 9.79 ± 4.09  | 1.51 ± 0.28           | 3.49 ± 2.08         | 4.79 ± 2.32      |
| 7    | Full              | 9.66 ± 4.19  | 1.23 ± 0.21           | 3.55 ± 2.17         | 4.89 ± 2.40      |
|      | Up to two sensors | 9.52 ± 4.21  | 1.24 ± 0.22           | 3.49 ± 2.28         | 4.78 ± 2.30      |
| 8    | Full              | 9.81 ± 4.36  | 1.30 ± 0.23           | 3.50 ± 2.13         | 5.01 ± 2.52      |
|      | Up to two sensors | 9.84 ± 4.10  | 1.49 ± 0.27           | 3.17 ± 1.96         | 5.18 ± 2.44      |
| 9    | Full              | 9.55 ± 4.11  | 1.34 ± 0.24           | 3.26 ± 2.00         | 4.96 ± 2.43      |
|      | Up to two sensors | 9.71 ± 4.11  | 1.38 ± 0.24           | 3.33 ± 2.01         | 5.01 ± 2.43      |
| 10   | Full              | 9.86 ± 4.27  | 1.43 ± 0.28           | 3.62 ± 2.29         | 4.82 ± 2.33      |
|      | Up to two sensors | 9.58 ± 4.20  | 1.42 ± 0.28           | 3.50 ± 2.24         | 4.66 ± 2.30      |

system state inferences (processed observations), conditioning on the system's core state. Note that an alternative approach, when real data are readily available, is to use observation sequences. That is, observation sequences (or synthetic sequences obtained using simulations) can be used in expectation maximization algorithms such as Baum-Welch algorithm (Bilmes et al., 1998) to estimate information matrices. Note that the information matrices may also be known *a priori* through system testing/sensor specifications. In such a case, we may forgo the proposed pre-processing step.

We define a set of mutually exclusive and collectively exhaustive intervals based on the system state means in different conditions. That is, each interval includes only one of the system state means,  $\mu_s$ . More specifically, we let  $\delta_s = \frac{\mu_s + \mu_{s+1}}{2}$ ,  $s \in \{0, \dots, \gamma\}$ . Consequently, we define the interval containing system mean in state  $s \in \{1, \dots, \gamma\}$ , as  $I_s = (\mu_s - \delta_{s-1}, \mu_s + \delta_s]$ . For the cases where the system is in brand new ( $s = 0$ ) and failed ( $s = \gamma + 1$ ) states, the corresponding intervals are  $I_0 = (-\infty, \mu_0 + \delta_0]$  and  $I_{\gamma+1} = (\mu_{\gamma+1} - \delta_\gamma, +\infty)$ , respectively. Recall that  $\tilde{X}_a^s$  denote the weighted average of sample means of the data collected from sensors under action  $a \in \{1, 2, \dots, 2^D - 1\}$  when the system is in core state  $s$  (Eq. (1)). Given that the system core state is  $s \in S$ , the probability that the observed system state is  $o$  ( $o \in \mathcal{O} \setminus \{-1\} = S$ ) is the probability that random variable  $\tilde{X}_a^s$  falls in the unique interval  $I_o$  that contains the mean of the observed system state  $\mu_o$ . Note that  $\bigcup_{s \in S} I_s = \bigcup_{o \in \mathcal{O} \setminus \{-1\}} I_o$ . Specifically, given the system core state is  $s$ , the probability that the observed system state is  $o$  can be calculated as follows.

$$Q_a(o|s) = P(\tilde{X}_a^s \in I_o | s, a) = \begin{cases} P(\tilde{X}_a^s \leq \mu_0 + \delta_0) = \Phi\left(\frac{\mu_0 + \delta_0 - \mu_s}{\sigma_s^a}\right), & o = 0, \\ P(\mu_o + \delta_{o-1} < \tilde{X}_a^s \leq \mu_o + \delta_o) \\ = \Phi\left(\frac{\mu_o + \delta_o - \mu_s}{\sigma_s^a}\right) - \Phi\left(\frac{\mu_o + \delta_{o-1} - \mu_s}{\sigma_s^a}\right), & o = 1, \dots, \gamma, \\ P(\tilde{X}_a^s > \mu_{\gamma+1} - \delta_\gamma) = 1 - \Phi\left(\frac{\mu_{\gamma+1} - \delta_\gamma - \mu_s}{\sigma_s^a}\right), & o = \gamma + 1, \end{cases} \quad (9)$$

where  $\Phi(\cdot)$  in the cumulative distribution function of the standard normal distribution.

## Appendix B. Cost breakdown for system with two sensors

Table 11 presents the cost breakdown of the optimal policies for the two sensor case under full action space,  $A$ , and restricted action space  $A_1^R$ . As seen in this table, when the action space is restricted to the use of only one sensor, the combination of sensors is not allowed. Consequently, this restriction generally leads to a reduction in data observation cost, albeit at the expense of decreased accuracy and increased sensor failure and/or replacement cost. Specifically, in cases such as baseline, IV, V, and XI, the restricted action space results in lower data observation costs. However, it is important to note that for these cases, the optimal policies under the restricted action space incur higher silent failure and/or replacement costs.

## Appendix C. Frequency of optimal sensor sets in system with ten sensors

Fig. 5 presents the frequency (in 1000 simulation runs) with which  $m$ -tuple sensor sets are used for monitoring under  $A$ ,  $A_2^R$ , and  $A_1^R$  for Cases 1–10. As seen in the figure, in Cases 1–10, under  $A$  where no restrictions are imposed on the optimal action set, the use of more than two sensors is not very common. Indeed, in only three cases (Cases 1, 3 and 8) more than two sensors at a time are used for system monitoring under  $A$ .

## Appendix D. Cost breakdown for system with ten sensors

Table 12 presents the cost breakdown of the optimal policies for the ten cases under full action space,  $A$ , and restricted action space with up to two sensors,  $A_2^R$ . As seen in this table, when restricting the number of sensors to be used, the combinations of prescribed sensors generally have higher accuracy to compensate for the imposed restriction. In some cases, this leads to higher data observation cost under the restricted action space; but this higher observation cost is compensated for with lower silent failure and/or replacement costs as more accurate sensors are used. For example in Cases 1, 2, 6, 7, and 8, higher data observation cost is incurred under the restricted

action space. For these cases, however, the optimal policies under the restricted action space incur lower silent failure and/or replacement costs.

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